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Technical Report No. 65 (Revised 2022)

Technology Transfer

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Technical Report No. 65 (Revised 2022)

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1.0 Introduction

Technology transfers play an increasingly important role in the regulated pharmaceutical and biopharmaceutical industries. The expanded use of contract development and manufacturing organizations, advances in manufacturing flexibility, increase in corporate mergers and acquisitions, and enhanced regulatory complexity all contribute to the need for technology transfers to maintain the continuity of drug supply. In order to replicate systems safely, maintain high-quality standards, and meet regulatory requirements, manufacturers have established approaches to successfully transfer manufacturing and analytical technology and knowledge from one site to another.

Pharmaceutical technology transfer (TT) uses a systematic approach to ensure the knowledge required to execute the manufacturing process and product testing performed at one site can be incorporated at a new site, while maintaining understanding and product characteristics. This results in the same safe, effective, and consistent pharmaceutical product being manufactured at both locations. TTs can occur at all stages of the product lifecycle, from drug development through clinical trials to commercial release.

The information in this technical report can be used to guide TT of both small- and large-molecule drug substances (biologics) as well as sterile and nonsterile drug and biological products. This document describes the steps for a single TT event. For large, complex TTs, these tools can be implemented in separate parallel processes, with a TT team established and a procedure executed for each discrete aspect being transferred. For instance, separate TT teams could be established for the transfer of drug substance manufacturing, drug product manufacturing, and packaging, with a management oversight committee ensuring the interconnectivity of the overall TT program across the supply chain.

The product realization requirements outlined in this technical report follow the guidelines of the International Council for Harmonization of Technical Requirements for Pharmaceuticals for Human Use (ICH) and should be documented in the chemistry, manufacturing, and controls (CMC) section of the regulatory filing.

1.1 Purpose

This technical report aims to provide a standardized approach to the TT process by supplying a matrixed "Reference Guide to Technology Transfer Activities and Deliverables," which can be used to coordinate cross-functional TT activities that help achieve operational readiness and culminate in regulatory approval. As specific details around each TT differ greatly, based on the type and complexity of the manufacturing process or analytical method being transferred, only a certain level of detail could be covered in a single technical report. This technical report is intended to provide a level of detail and approaches that can be applied across TT types.

1.2 Scope

This technical report covers the full range of requirements needed for TTs, such as new product introductions and the transfer of products that are being manufactured for phase III clinical trials or commercial release, which are the most comprehensive in nature. In transfers of products at an earlier stage of development, when process and product knowledge is less understood and therefore CMC requirements are less detailed, certain steps described in this technical report will not be necessary. This report does not detail those differences.

While this information was not developed specifically for medical devices, combination products, and advanced therapy medicinal products due to their unique requirements, many of the fundamentals expressed here can be applied to those products.

For this technical report, it is assumed that all involved parties have established systems for commercial production. Thus, standard requirements such as equipment and facility maintenance, annual product

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reports, lifecycle validation, complaint handling, and continued process verification (CPV) programs are not addressed. For purposes of this technical report, the TT ends with the handover to operations upon regulatory approval.

2.0 Glossary and Abbreviations

Definitions have been provided to help clarify the concepts discussed in this technical report. While some of the definitions vary among companies and regulatory agencies, the definitions described below are consistent for use within this document. Where a definition is taken verbatim from another source, the source is cited.

Consumables
Items (e.g., storage bags, tubing, filters, diaphragms, flasks) that form or are a part of process equipment and are replaced regularly.

Critical Process Parameter
A process parameter whose variability has an impact on a critical quality attribute and therefore should be monitored or controlled to ensure the process produces the desired quality ***(1)***.

Engineering Batch
A manufacturing run performed to demonstrate the transfer of a manufacturing process and confirm the suitability of qualified equipment, controls, and analytical methods that are appropriate to the lifecycle phase of the product. This may also be referred to as an engineering run, demonstration batch, or demonstration run.

Facility Fit Assessment
An assessment performed on the manufacturing capabilities of the receiving unit during a TT. This assessment will consider equipment, design, capacity, and risks associated with manufacturing the product being transferred to the receiving unit's facility.

Handover
The stage of a TT in which the TT team transitions the product's manufacturing process to the post-transfer operations team who will be responsible for the ongoing manufacturing of the product at the receiving unit. This stage may also be referred to as commercial or project turnover or hand-off.

Knowledge Transfer
A practical, systematic approach to transfer the body of knowledge available for a specific product and process, including critical-to-quality product attributes, process parameters and control strategy, process capability, testing, and manufacturing within a quality system's infrastructure between two locations.

Knowledge Management
A systematic approach to acquiring, analyzing, storing, and disseminating information related to products, manufacturing processes, and components ***(2)***.

Quality Target Product Profile
A prospective summary of the quality characteristics of a drug product that ideally will be achieved to ensure the desired quality, taking into account safety and efficacy of the drug product ***(1)***.

Stage Gate
A review meeting conducted during the process of TT to verify that the requirements for the current stage have been completed and that the team is ready to move into the next stage of the TT project. The meeting typically is led by the TT team and TT project manager, who present findings, results, and potential and identified risks with proposed mitigation strategies, decisions, and next steps to the steering committee for the TT. If risks are not acceptable or certain requirements have not been met, the steering committee may identify additional actions that must be performed before proceeding to the next stage. A stage gate also may include a review of lessons learned. This may also be referred to as a milestone review meeting.

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User Requirements Specification (URS)
The set of owner, user and engineering requirements necessary and sufficient to create a feasible design meeting the intended purpose of the system.

2.1 Abbreviations

APR Annual product report

CAR Capital appropriation request

CDMO Contract development and manufacturing organization

CMC Chemical, manufacturing, and controls

CMO Contract manufacturing organization

EHS Environment, health, and safety

FAT Factory acceptance test

GMP Good manufacturing practices

KM Knowledge management

KT Knowledge transfer

MAH Marketing authorization holder

PAI Preapproval inspection

PM Project manager

PPQ Process performance qualification

QTPP Quality target product profile

RU Receiving unit

SAT Site acceptance test

SME Subject matter expert

SU Sending unit

TT Technology transfer

URS User requirements specification

VMP Validation master plan

3.0 Types of Technology Transfers

ICH *Quality Guideline Q10: Pharmaceutical Quality Systems* describes TTs as new product transfers during development through manufacturing and transfers within or between manufacturing and testing sites for marketed products ***(3)***. This broadly encompasses all transfers, including analytical method transfers to an alternate laboratory, transfers to scale up a development product, and transfers from the laboratory to a current good manufacturing practices (GMP) production environment. When establishing a TT program, it is important to understand the impact of the product lifecycle phase, the type of sites involved, and the associated CMC requirements.

Although this section describes all types of TTs, this technical report focuses on the fundamental activities necessary for introduction of a new product to a manufacturing site when the product is moving from clinical phase III to commercial manufacturing or from one commercial manufacturing site to another.

3.1 Product Lifecycle

During its lifecycle, a product typically progresses through discovery and development, culminating in commercial readiness and regulatory approval as depicted in **Figure 3.1-1**. TTs are a key part of

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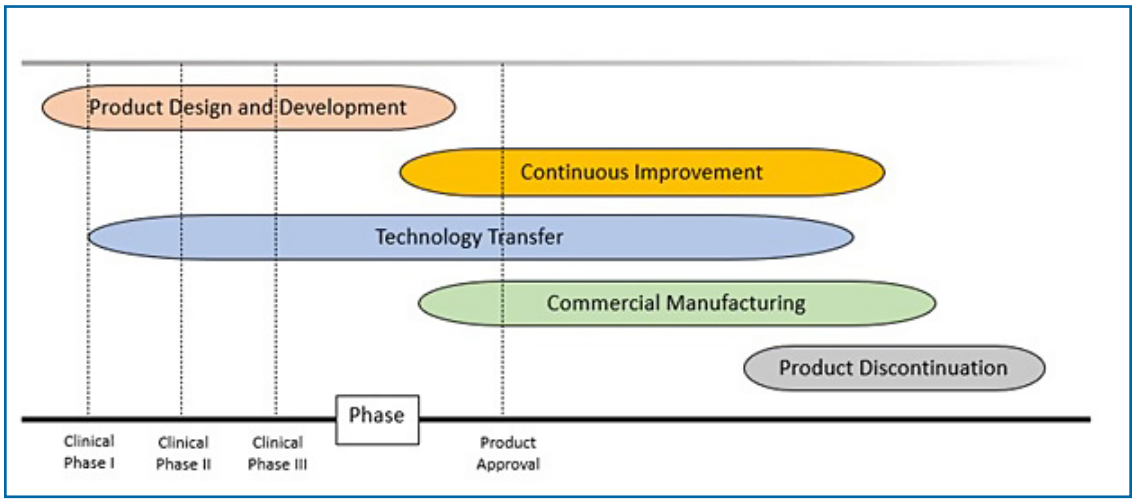

Figure 3.1-1 Pharmaceutical Product Lifecycle

the lifecycle as a new drug product and its corresponding drug substance move from clinical phase I through to commercial manufacturing, often at different production scales and different manufacturing sites. While TTs may occur at any point throughout the product lifecycle, each TT should have its own purpose (see **Section 4.3)** and will have a discrete beginning and end.

With the aim of driving reproducibility, elements of TT are performed at each phase of the product lifecycle as knowledge of the process, product, and testing increases. TTs can be described based on product lifecycle phase: transfers within the clinical phases (clinical-clinical); transfers from the development phase, which may be clinical phase III, to the commercial phase (development-commercial); and transfers within the commercial phase (commercial-commercial). This section further describes each of the lifecycle phase groups and their influence on TT requirements.

3.1.1 Clinical-Clinical Technology Transfers

While this technical report does not describe the requirements for clinical-clinical TTs, TTs at clinical stages should be developed, reviewed, and executed with thoughtful consideration and should be phase-appropriate from a CMC perspective. TTs during these phases should address clinical patient safety and begin building the foundation for future commercial production including the development of the TPP. During early clinical phases, the involved sites should constantly consider future needs related to scaling up and validating the manufacturing process, determining the type of facility required, and defining the product, process, and analytical characteristics necessary to ensure the quality, safety, and efficacy of the product being transferred. Clinical phase TTs often are complicated by small material demands and by extended pauses for site inspections and regulatory approvals as clinical data is gathered, making knowledge management throughout the early clinical phases especially important.

3.1.2 Development-Commercial Technology Transfers

Development-commercial TTs, which often begin when the product is in clinical phase III, involve the transfer of the full scope of CMC requirements. By clinical phase III, the product, process, and analytical development should progress in parallel with the TT program to define the final CMC requirements for commercial readiness and process and analytical validation. For instance, the final dosage form and dosage should be known; the drug substance manufacturing process should be established such that there are minimal changes in the impurity profile from clinical phase III to process validation; process parameters and critical quality attributes (CQA) should be defined, including characterization of critical components and materials; and initial stability for both drug substance and drug product should have been evaluated. Risk assessments should be performed throughout final development to determine the impact areas for commercial manufacture. PDA *Technical Report No.*

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54: Implementation of Quality Risk Management for Pharmaceutical and Biotechnology Manufacturing Operations provides more information on quality risk management and the performance of risk assessments ***(4)***. The objective of the TT activity is to establish comparability between the SU and RU processes or methods and, as a result, provide the regulatory authorities the assurance needed to allow the use of the RU to supply clinical material or commercial inventory into their market.

3.1.3 Commercial-Commercial Technology Transfers

In commercial-commercial TTs, the product is already being marketed and produced at an existing facility, yet business may dictate the need to move or expand production of the drug substance or drug product to another facility. This may require an operational scale-up or scale-down. The addition of a new market may require the business to use alternate packaging facilities, which would require TT of the packaging operations. Alternate laboratories also may be used, requiring transfer of the analytical methods to the new location. In TTs involving a marketed product, the original or current manufacturing location should provide the full CMC requirements, with historical product and process knowledge, that demonstrate manufacturing robustness. These types of TTs may be an opportunity to implement process improvements at the receiving unit, provided that those improvements align with the business and regulatory strategies for the TT.

3.2 Site Types

Typically, two types of sites are involved in a TT: the sending unit (SU) and the receiving unit (RU). The SU is the site with the current knowledge of the product, process, and analytical methods, while the RU is the location where the product will be manufactured or tested in the future. The use of contract manufacturing organizations (CMOs) and contract development and manufacturing organizations (CDMOs), the advent of the virtual company business model, and the rise of global manufacturing supply chains have expanded the variations of SU and RU sites involved in TTs.

Both the SU and RU sites can be classified as internal or external. Internal sites include any facility for which the existing or future marketing authorization holder (MAH) has ownership or complete control, while external sites are separate from the MAH and include CMOs, CDMOs, business partners, and external laboratories. The type of site and its relationship to the MAH impacts the effectiveness of knowledge transfer as well as the overall TT team and governance framework described in **Section 4.0**.

3.2.1 Internal-Internal Technology Transfers

A TT performed between locations of the same company or facility is considered the most manageable in terms of knowledge transfer between the SU and RU sites for several reasons. Most likely, these sites are governed by the same quality management system (QMS), have integrated document management systems, and employ a standardized TT process. If the company is transferring technology for an existing commercial product, historical data on process or test performance also should be readily available. Even with use of the same site, however, a complex process or an analytical method that draws on unique knowledge held by the SU can still present challenges.

If the internal SU and RU sites are in different geographical locations, an initial evaluation will be needed to determine the potential impact of language, local implementation of the global QMS, import and export requirements, and regulatory requirements. These TTs typically will require additional budget for site-to-site interactions that allow the SU to share knowledge with the RU and participate in initial manufacturing at the new location.

3.2.2 Internal-External or External-Internal Technology Transfers

The MAH may choose to work with an external company in many different scenarios, such as:

- Product developed internally will be manufactured at commercial scale at an external company, such as a CMO or CDMO, for cost or time-to-market purposes or due to limited capability in internal facilities.

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- Product developed at a CDMO will be manufactured at an internal site.
- Commercialization was performed internally and transferred to an external company after receiving regulatory approval, or vice versa.
- Use of multiple manufacturing facilities will better meet demand and/or provide risk mitigation for supply.
- Expansion of a product to a new market requires alternate global manufacturing capability.
- A corporate acquisition or merger results in manufacturing consolidation.

In these types of scenarios, the TT project manager (PM) has the additional challenge of managing the TT activities across multiple sites, organizations, and cultures.

Unlike internal-internal transfers, any transfer to or from an external site may involve differences in organization, quality systems, and specific procedures. In some cases, the external organization may have competing priorities, which can be difficult to discern. Defining the differences in organizational structure, quality systems, and business priorities early can facilitate alignment on the TT boundaries and success factors. Agreements on the presence of SU subject matter experts (SMEs) at the external site during the TT process to ensure a successful knowledge transfer is an important point that cannot be overlooked and should be documented in the appropriate agreements.

3.2.3 External-External Technology Transfers

Often, the MAH must manage a transfer between two external sites, which may also involve two different companies. This type of TT can present the greatest challenge, particularly if the two external parties are not known to each other (e.g., the result of a business strategy). If the external parties cannot interact, the MAH will need to act as the SU for the TT and facilitate the knowledge transfer to the RU. Agreements on the presence of MAH and SU SMEs at the external site during the TT process to ensure a successful knowledge transfer is an important point that cannot be overlooked and should be documented in the appropriate agreements.

4.0 Technology Transfer Process and Oversight

Regulatory authorities expect companies to have a documented TT process that describes the steps to be taken and the compliance requirements for each step.

Figure 4.0-1 illustrates a TT process using six stages and stage gates. These six stages and stage gates, outlined briefly in this section, are discussed in more detail in **Section 5.0,** and their use is described in the "Reference Guide to Technology Transfer Activities and Deliverables" (**Figure 5.0-1**).

Each stage of a TT serves a purpose and has identifiable outcomes. At each stage, stage gates allow a steering committee to determine whether those outcomes have been met and to decide if the TT should proceed (see **Section 4.2**).

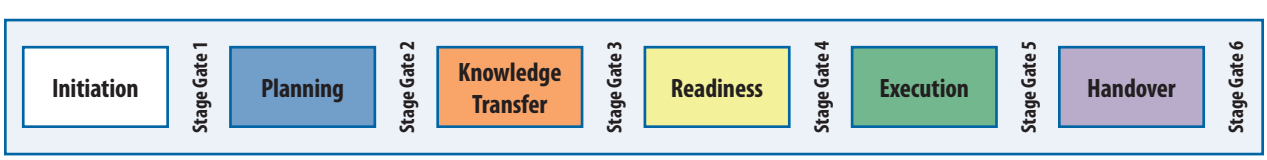

Figure 4.0-1 The Six-Stage Technology Transfer Process

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In the Initiation stage, potential RUs are evaluated to identify a site with a risk profile that meets the business objectives. Once a suitable RU is selected, a TT Team is formed at the RU site. In the Planning stage, the TT Team assesses the TT risks and identifies elements critical to the knowledge transfer and overall timing. In the Knowledge Transfer stage, the formal transfer of the detailed requirements specific to the TT begins. Once the knowledge transfer elements are obtained, the TT Team works in the Readiness stage to perform the activities and achieve the deliverables required to achieve full operational readiness. In the Execution stage, the RU demonstrates and qualifies the process at scale, culminating in a regulatory submission. The final Handover stage closes the TT with regulatory approvals and allows normal operations to begin.

Each TT involves unique technical, business, and regulatory requirements that should be considered in structuring the project. Each company may choose to add or consolidate stage gates and TT process steps to fit the type of TT being undertaken, the company's organizational structure, and its level of risk tolerance.

In every TT, assessing and managing risks is essential. A mature quality risk management program, like that described in PDA TR 54, should be in place prior to starting a TT project.

4.1 Cross-Functional Roles and the Project Manager

A successful TT requires the engagement of experts from cross-functional areas as well as strong project management. **Table 4.1-1** outlines the basic functional roles that maintain responsibilities throughout a TT and should be represented on a TT team. Because each company is organized differently, the labels for these functions may differ from one organization to the next.

More detailed information about these roles is provided in **Section 5.0,** and the "Reference Guide to Technology Transfer Activities and Deliverables" (**Figure 5.0-1)** offers an overview of the roles of each function at each stage of the TT.

Table 4.1-1 TT Cross-Functional Roles

Function	Description
Business	Strategy, marketing, finance, legal, sales
TT PM	Leads the overall TT effort. There also should be PMs at each involved site to support the TT
Process	Process chemistry and engineering, process performance qualification, validation[1]
Analytical	Raw material, in-process controls, product, stability (development and quality control)
Engineering	Capital installation and qualification, equipment maintenance
Manufacturing	Plant operations and EHS (environment, health, and safety regulations)
Regulatory	Health authority strategy and filing
Quality	QMS requirements, inspection readiness (i.e., PAI)
Sourcing/Supply Chain	Material sourcing, production planning, logistics

[1] Some companies have a standalone group for validation, but this technical report discusses validation within the identified functions.

Once the decision has been made to proceed with the TT, all involved sites should document expectations, roles, and responsibilities in a formal TT agreement. The TT agreement also should specify the measuring units to be used during the TT (e.g., for mass, temperature, and volume). If the transfer is internal-internal, a shared procedure may be developed in place of a TT agreement. This TT agreement or shared procedure provides information about the level of involvement of each functional role at the SU and RU that will allow each site to resource a TT team.

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Each site involved in the TT should develop a cross-functional team to execute the TT. Each team should include a lead person who is an SME for each functional role described in **Table 4.1-1.** A single person may serve as lead for more than one functional area. While each site should have a team, references to the "TT Team" in this document mean the team at the RU unless otherwise indicated.

Team composition may vary slightly for each TT and at each involved site. In some cases, a single TT Team may be formed at the RU, with the MAH or SU also assigning staff for each functional area to provide oversight and facilitate accurate and efficient knowledge transfer. In other cases, parallel teams may be established at each site. In an external-external TT, each external location should have its own cross-functional TT team. Roles also will vary at each site. For example, the MAH and RU Regulatory functions may have more significant roles than the SU Regulatory function.

The TT team at each site should be involved at nearly every stage throughout the process. The TT team at the MAH and/or SU will play a significant role during the Initiation, Planning, and Knowledge Transfer stages, but should be engaged through to the Handover stage. This TT team will ensure the RU has the information necessary to manufacture the product and will provide expertise and learnings throughout the TT process.

The TT Team at the RU will have primary responsibility for implementing the TT. This team should be involved from the Planning stage, although their role will become more significant as the TT progresses. The TT Team at the RU is responsible for ensuring that their facility meets the project scope, business needs, and site requirements to manufacture the new product.

Each site involved in the TT should assign an individual to serve as team lead or PM for the site. This role requires basic technical knowledge across the functional areas as well as knowledge of CMC requirements. The person serving as team lead may or may not have a "project manager" job title but serves as a primary point of contact and facilitates communication, knowledge transfer, and task completion. In internal-internal TTs, where the SU and RU are located in the same facility (i.e., the product is being transferred from one production line to another), a single team lead may be adequate. The team lead may be chosen from one of the functional areas involved in the TT, could be in management, or could be a contractor.

One PM must be the primary coordinator for the project. This technical report refers to this primary coordinator as the TT PM. Often, the PM at the RU will serve as the TT PM. In an external-external TT, however, an individual from the MAH may act as the TT PM to ensure a smooth and continuous flow of knowledge transfer. In large, complex TTs, the TT PM may be someone different from the site PMs. Wherever the TT PM is located, this role is critical in facilitating constant communication and should be the primary contact point for changes, decisions, and team alignment.

4.2 TT Management Oversight

The organizational structure supporting a TT requires clarity and definition to ensure the TT Team can maintain focus, manage resourcing, and resolve issues. A defined structure for the TT allows for improved communication and clear definition of priorities among team members and management. A TT is usually considered a special project with a fixed scope and defined goals. As such, members of the TT Team may not be permanently assigned to the TT and may continue to report functionally outside of the team. TT Team members typically are assigned to the team under the direction of the TT PM for the duration of the project.

Figure 4.2-1 depicts a structure for a TT project that aligns with the TT process described in this technical report.

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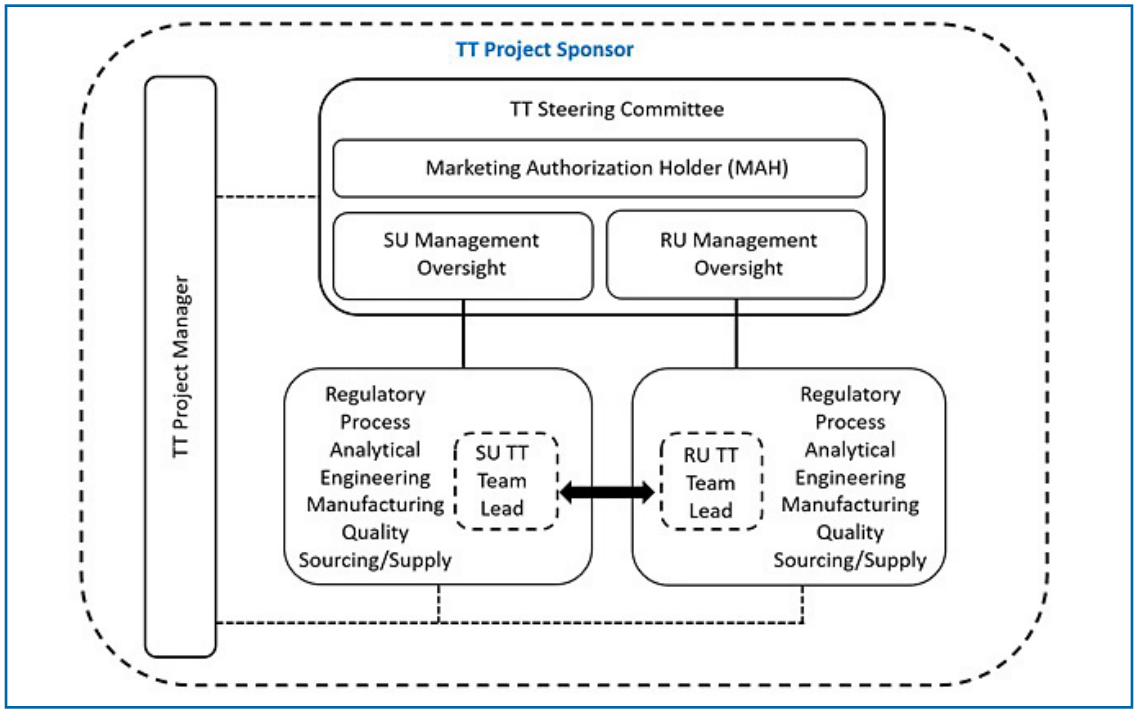

Figure 4.2-1 Typical TT Project Structure

As the key stakeholder accountable for the product with regulatory authorities, the MAH determines the goals for the TT and is the contract giver.

Once the decision has been made to proceed with the TT, a steering committee is formed. Each site involved in the TT should assign steering committee members who will be accountable for achieving the TT goals. Typically, steering committee members are one management level above the TT team for each of the functional areas working on the TT and have decision-making authority. The steering committee is empowered to remove roadblocks, assume risk, provide resources, and determine whether the requirements of each stage have been fulfilled.

Upper management may identify a Project Sponsor to oversee the TT. The person in this role typically is the single point of contact between the TT and upper management and has overall accountability to ensure the TT is delivered as required by the business. The Project Sponsor may be part of the steering committee, but the steering committee is accountable to the Project Sponsor.

Stage gates at each of the six stages allow the steering committee to determine whether a TT is ready to proceed to the next stage. Often TT activities will be performed across stages in parallel rather than sequentially due to business needs. Each stage gate, however, allows the steering committee to ensure that all necessary activities and deliverables for that stage have been satisfactorily completed and that the TT charter, as described in **Section 4.3**, can be accomplished. When a TT Team finds information during a stage that may change the TT boundaries and/or assumptions, it should be communicated immediately to the steering committee rather than waiting for the stage gate review.

4.3 TT Charter and Success Factors

The overall framework of a TT should be documented in a charter. The charter should be developed by the TT Team and must be approved by the steering committee and, if applicable, the Project Sponsor. Approval by the steering committee establishes that the sites have reached the alignment necessary to proceed.

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The charter typically includes the following information:

- Definition of TT completion and success factors for the TT
- Business assumptions, major risks, and decisions
- Target markets and market volumes
- Regulatory strategy
- Steering committee members
- Relevant stakeholders and how they are involved and connected to the project
- PMs and functional points of contact at SU and RU, if needed
- Communication plan
- Timeline with critical path objectives
- Lead SMEs for each cross-functional area
- Estimate of resource requirements

The success factors defined in the charter will serve as the boundaries within which the TT Teams will work (see **Section 5.1.2** for more details). These success factors should be developed by the TT PM with agreement and input from the steering committee. Often based on business and regulatory strategic objectives, some typical success factors that would be documented in the TT charter might include:

- Right-first-time execution of process validation batches
- Demonstration of comparability
- Regulatory filing submitted
- Preapproval inspection (PAI) authorized
- Budget goals achieved
- Time-to-market requirements met
- Handover to the RU post-transfer operations team
- Process improvements impacting production efficiency or cost

Any changes to the requirements for TT success or the business assumptions defined in the charter should prompt the steering committee to revisit the Initiation and Planning stages to assess the potential impact of the changes.

Any changes to the charter after the Planning stage should be approved by the steering committee and documented.

5.0 Using the Reference Guide to TT Activities and Deliverables

The "Reference Guide to Technology Transfer Activities and Deliverables," provided in its complete form in **Figure 5.0-1**, uses a matrix approach to show the recommended activities and deliverables for which each functional area is responsible at each stage of the TT. To ensure effective knowledge management of the CMC requirements for the product being transferred, each stage of the TT process will have defined deliverables for each of the functional areas. While the Reference Guide recommends a specific functional area to lead each activity and deliverable, all functional areas must intersect and work in parallel to ensure success of the TT.

The sections below discuss the key elements of the Reference Guide for each TT stage and summarizes common activities and deliverables for each stage, organized by functional area. A table of key considerations is provided at the end of the discussion for each stage, and case studies demonstrating the use of the process and the Reference Guide are provided in **Section 8.0** (**Appendix**).

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Figure 5.0-1 PDA Reference Guide to Technology Transfer Activities and Deliverables

STAGE	Business	TT PM	Process	Analytical	Engineering	Manufacturing	Regulatory	Quality	Sourcing/Supply Chain
Initiation	***Activities***								
	Develop Business Case	Coordinate Subject Matter Expert (SME) Assessments	Determine Preliminary Process Flow	Perform Initial Method Evaluation	Perform Permit Reviews (Licenses, Construction)	Perform Permit Reviews (Environment, Health, and Safety (EHS))	Evaluate Market Requirements	Perform RU Quality Assessment(s)	Perform RU Capacity Analysis
	Identify Technology Transfer (TT) Project Manager	Perform Initial Risk Assessment	Define Batch Size	Evaluate Laboratory Capabilities	Evaluate SU and RU Site Capabilities	Assess EHS Requirements	Develop Regulatory Strategy		Evaluate Import/ Export Requirements
	Perform Patent Review	Perform Due Diligence	Develop/Obtain Sending Unit (SU) Qualtiy Target Prodcut Profile (QTPP) and Product Specifications		Estimate Capital Requirements				Evaluate Material Sourcing
	Develop Product Strategy	Drive Receiving Unit (RU) Selection Decision	Evaluate Manufacturing Process Equipment Requirements						Determine Requirements for Tightly Regulated Materials
	Obtain Non-Disclosure and/or Confidentiality Agreements	Develop/Update Project Plan	Identify Scale-Up Needs						
	Deliverables								
	Business Case	Initial TT Charter	Process Flow/ Description	Laboratory Evaluation Report	Facility Fit Assessment	EHS Viability Report	Regulatory Strategy	Quality Assessment(s)	Preliminary Product Supply Chain Strategy, Including Map
	Product Strategy	High-Level Project Plan	Process Fit/Gap Assessment	Methods Evaluation Report	Permit Requirements	Permit Requirements	Regulatory Approval Timing		Critical Material List
	Patent Review Report	RU Due Diligence Report	Product QTPP		Capital Estimate				
	Non-Disclosure and/ or Confidentiality Agreements	Risk Assessment Report							
	RU Site Selection								
	Steering Committee Formed								

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(Figure 5.0-1 Continued)

STAGE	Business	TT PM	Process	Analytical	Engineering	Manufacturing	Regulatory	Quality	Sourcing/Supply Chain
Planning	***Activities***								
	Approve Resource Needs	Define Resource Requirements	Detailed Process Risk Assessment	Detailed Analytical Risk Assessment	Define RU Requirements	Operational Requirements Analysis	Obtain Current Dossiers	Perform RU Quality Audit	Production Planning
	Approve Capital Plan	Define Critical Path Items	Align QTPP and Product Specification with RU	Define Analytical Equipment Requirements	Develop Capital Plan	Perform EHS Assessment	Interact with RU Regulatory Affairs Function	Define SU/ RU/Marketing Authorization Holder (MAH) Quality Management System (QMS) Responsibilities	Determine Product Logistics
	Interact With Customers	Develop Detailed Project Plan	Define Material Handling Requirements	Determine External Lab Needs	Write Capital Appropriation Request (CAR)	Initiate Permits	Interact with Health Authorities if Necessary	Define Documentation Requirements	Source Materials/ Disposables/ Packaging/ Consumables
	Sponsoring Steering Committee	Create Communication Plan	Define Commercial Process at RU	Define Sampling Plan	Write Change Controls			Define Comparability Requirements	Order Long-Lead Materials
		Implement Documentation Sharing	Define Bill of Materials					Develop Stability Plan	
		Hold Steering Committee Meetings							
	Deliverables								
	RU Supply Agreement	Final TT Charter	TT Change Control	Analytical Risk Assessment Report	Capital Project Plan	Operational Requirements Report	Current Product Filings and Annual Product Reports	RU Audit Report	Product Supply Chain Strategy
	External Lab Supply Agreement	Project Plan	Process Risk Assessment Report	Sampling Plan	Equipment/Facility Change Controls	EHS Assessment Report	Current Compendial Specifications	Quality Agreements SU/ RU/MAH/External Labs	Plant Schedule Availability
	Resource Plan	TT Agreement between SU/RU	Commercial Process Established		CAR Approval		Comparability Protocol	TT QMS	Long Lead Material Plan
	Stage Gate Approval	Critical Paths Identfied	Initial Bill of Materials		Change Controls			Quality Improvement Plan	
		RACI	Comparability Plan					Stability Plan	
		Communication Plan							
		Final TT Team Formed							
		Kickoff Event							

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(Figure 5.0-1 Continued)

STAGE	Business	TT PM	Process	Analytical	Engineering	Manufacturing	Regulatory	Quality	Sourcing/Supply Chain
Knowledge Transfer	***Activities***								
	Removes Roadblocks	Communicate with SU	Define Critical Process Parameters	Evaluate Material and Product Method Requirements	Identify Plant/ Analytical Equipment Requirements	Assess SU Master Batch Record	Obtain SU Product Label	Perform Quality Risk Assessment	Materials/Disposables/ Packaging Supply Agreements
	Executes Resource Plan	Obtain Process Knowledge from SU	Determine In-Process Tests	Evaluate In-Process Test Methods	Identify Disposables	Define Personal Protective Equipment (PPE) Requirements	Refine Regulatory Strategy	Perform Material Supplier Qualifications	Obtain Material Specifications
	Configure Material Requirements Planning/Enterprise Resource Planning Systems for New Product Introduction	Facilitate Writing of RU Specifications	Finalize Process at RU	Evaluate Cleaning Methods	Write Capital Validation Master Plan	Define Operations Documentation	Provide Regulatory Support	Develop Training Curricula	Source Reference Stds
			Define Cleaning Procedure and Limits	Evaluate Microbial Methods	Design/Order Capital Items		Review Stability Plan	Establish Expiry/ Retest Date	Develop Bill of Materials
			Assess Rework/ Reprocess	Order and Qualify Reference Standards	Design Environmental Disposal				
			Write Validation Master Plans (VMPs)	Assess Compendial Methods	Write Change Controls				
			Define Process Monitoring Requirements	Define Harmonized Methods					
			Demonstrate Process						
	Deliverables								
	Stage Gate Approval	SU Technical Package	RU Technical Package	Reference Standards	User Requirements Specification (URS)	PPE Requirements	Updated Regulatory Strategy	Quality Risk Management (QRM) Report	SU Material/Packaging/ Consumables Specifications
	Change Orders	RU Material Specifications	Mass Balance	Analytical Method Validation Plan	Equipment Validation Master Plan		Health Authority Interaction	Supplier Qualification Reports	Bill of Materials
		RU Product Specification	Waste Characterization		Engineering Drawings		Stability Plan	Compliance Training	Qualified Supplier List
			Process VMP		Change Controls				
			Cleaning VMP						
			Lab Demonstration Runs						

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(Figure 5.0-1 Continued)

STAGE	Business	TT PM	Process	Analytical	Engineering	Manufacturing	Regulatory	Quality	Sourcing/Supply Chain
Readiness	***Activities***								
	Completes Resource Plan	Write TT Protocol	Develop Control Strategy	Write Method Validation/ Transfer Protocols	Perform Factory Acceptance Test (FAT)/Site Acceptance Test (SAT)	Write Master Batch Records	Review Product Labeling Mock-Ups	Review and Approve Documentation	Order Materials
	Communicate Changes to Business Strategy	Remove Roadblocks	Demonstrate Process Monitoring	Execute Method Validations/Transfers	Install Capital Items	Write Standard Operating Procedures (SOPs) and Work Instructions	Review Change Order Notifications	Write Procedure for Batch Release	Define Distributors
	Establish Customers	Facilitate Decisions	Write Process Validation Protocol	Confirm Compendial Method Viability	Set Up Control System	Perform EHS Risk Assessment	Develop Submission Tracker	Train Quality Assurance Personnel on Process	Set Up Inventory Control Software Requirements
			Write Cleaning Validation Protocol	Write Stability Protocols	Write Capital Qualification Protocols	Perform Operations Training	Identify Necessary Permits & Licenses	Release Materials	
			Establish Hold Points	Perform Analytical Training	Perform Equipment/ Facilities Qualifications	Perform Pre-Cleaning		Audit Vendors and Suppliers	
			Develop Sampling Plan		Write Preventative Maintenance Plan	Set Up Warehouse		Develop Preapproval Inspection (PAI) Readiness Plan	
			Write Shipping Validation Protocols		Set Up Spare Parts Requirements	Execute Cleaning from Installation		Review Product Labeling Mock-Ups	
			Perform Process Training						
	Deliverables								
	Resources Available	TT Protocol	Control Strategy	Validated or Qualified Methods	Installed Equipment	Master Batch Record	Final Regulatory Strategy	Approved Documentation	Supply Chain Map
	Current Business Strategy		Process Validation Protocol	Qualified Reference Standards	Capital Qualification Protocols	SOPs and Work Instructions	Submission Tracker	Equipment Readiness Approval	Materials Delivery
	Customer Contracts		Cleaning Validation Protocol	Stability Protocols	Capital Qualification Reports	EHS Risk Register	Internal Product Labeling Approval	Deviation/Corrective Action/Preventive Action (CAPA) Closure	
	Stage Gate Approval		Hold Points	Trained Personnel	Preventative Maintenance Program	Trained Personnel		Qualified Vendors and Suppliers	
			Sampling Plan		Spare Parts List	Clean Equipment		PAI Readiness Plan	
			Shipping Validation Protocol					Internal Product Labeling Approval	

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(Figure 5.0-1 Continued)

STAGE	Business	TT PM	Process	Analytical	Engineering	Manufacturing	Regulatory	Quality	Sourcing/Supply Chain
Execution	***Activities***								
	Develop Launch Strategy	Ensure All Good Manufacturing Practices Documentation Is Complete	Oversee Production Runs	Perform Materials Testing	Support Process Performance Qualification (PPQ)	Perform Demo/ Engineering Batch	Coordinate Regulatory Filing	Assess Deviations/ CAPAs	Manage Product Storage Requirements Prior To Release
		Participate in PAI Audit	Investigate Deviations	Perform In-Process Testing	Write Deviations	Execute PPQ Batches		Review Materials Data	
			Gather/Analyze Process Data	Perform Batch Release Testing		Execute Cleaning from PPQ		Review Batch Data	
			Perform Shipping Validation	Perform Cleaning Testing				Review Documentation	
				Package Stability Samples				Lead Execution of PAI Readiness Plan	
				Test Demo/ Engineering Batch					
	Deliverables								
	Stage Gate Approval	Equivalency Study	Process Validation Report, VMP Closure	Approved Test Results	Deviation Reports/ CAPAs	Executed Batch Records	Submit Regulatory Filing	Materials Release	Prepare Release & Distribution Plan
	Approval for Regulatory Filing	TT Report	Cleaning Validation Report	Stability Program		Clean Equipment		Product Release	
	Launch Strategy		Deviation Reports/ CAPAs			Store Product		Deviation/CAPA Resolution	
			Cycle Time, Yield					Validation Documents Approval	
			Shipping Validation Report					Close Change Controls	
								PAI Approval	

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(Figure 5.0-1 Continued)

STAGE	Business	TT PM	Process	Analytical	Engineering	Manufacturing	Regulatory	Quality	Sourcing/Supply Chain
Handover	***Activities***								
	Determine Changes Needed for Commercial Production	Coordinate Knowledge Transfer to Post-Transfer Operations Team	Set up Continued Process Validation (CPV)	Maintain Stability Program	Define Changeover Requirements for Future Production	Knowledge Transfer to Post-Transfer Operations Team	Obtain Regulatory Approval	Monitor Stability Data	Validation Batch Distribution
		Document Lessons Learned in TT	Document Potential Process Improvements			Maintain Validation Batch Storage	Consider Post-Approval Changes	Set up Quality Management Review Requirements	Coordinate Transportation Strategy
	Deliverables								
	Commercial Resourcing to Implement Changes	Full Handover of Product	CPV Program, Report Template (or initial report when sufficient batches produced)	Stability Results/ Report	Product Changeover Plan	Trained Personnel	Regulatory Approval	Commercial QMS	
	Stage Gate Approval							Lead Quality Management Reviews	
								Commercial Quality Agreement	

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While a TT PM can use the Reference Guide to develop a robust TT program, the guide does not address each unique requirement for an individual product transfer. The TT Team will need to determine the specific requirements necessary to ensure that the RU obtains and implements all the knowledge needed to manufacture safe, effective, and consistent product in compliance with CMC requirements.

5.1 Initiation Stage

The Initiation stage covers the process of determining whether the TT of CMC activities for a pharmaceutical or biopharmaceutical product should be performed. In this stage, the team evaluating the need for the TT often has limited information on the product characteristics, process requirements, and analytical testing from the SU, particularly if the product is still under development. While the Initiation stage is primarily driven by the Business function of the MAH, technical experts also should conduct high-level risk assessments.

A PM should be assigned to facilitate RU site selection and to develop the initial charter. This stage results in a decision on whether to perform the TT, the choice of an RU site, the formation of a steering committee, and the development of an initial TT charter as well as an overall product strategy, including business and regulatory considerations. **Table 5.1-1,** excerpted from the Reference Guide (**Figure 5.0-1),** shows the activities and deliverables for the Initiation stage.

5.1.1 Business

The Business function at the MAH takes the lead role in the Initiation stage, developing the business case for initiating a TT activity. It facilitates obtaining key information from the SU, such as the quality target product profile (QTPP) and/or specifications to identify the product characteristics and high-level process requirements. Using a high-level risk assessment, the Business function develops the product strategy to guide the TT decision. Legal counsel also is engaged for intellectual property review to ensure the freedom to operate, if needed.

At this point, the RU site selection process can begin. If the product will be sourced externally, the Business function secures nondisclosure agreements or confidentiality agreements with potential RUs. The criteria laid out in the QTPP or product specification, the product strategy, the intellectual property review, and the information required for product lifecycle phase (see **Section 3.1**) are used in the site selection process. Typically, potential RUs are assessed for capability, age of facility, location, capacity, resourcing, costs, tax implications, compliance history, regulatory strategy, and other logistical considerations. Once a RU site is selected, the initial TT charter is approved, and the steering committee is formed. **Section 8.1** presents a case study of best practices and benefits of establishing a steering committee for a project.

The stage gate for Initiation results in a decision to proceed with the TT; selection of the RU, often with a signed contract; and approval of the initial charter. Two case studies, **Section 8.4** on using a regulatory strategy for decision-making and **Section 8.8** on TT of an aseptic drug product, illustrate factors to consider when selecting a RU or supplier.

5.1.2 Initiation Stage Project Manager

Involvement of a PM during the Initiation stage of a TT is critical to provide overall leadership and accountability for the project. The PM needs to be knowledgeable across all relevant functional areas and able to communicate across organizational levels. Additionally, this role requires such soft skills as facilitation, negotiation, decision-making, and motivation to effectively lead a cross-functional team to achieve business goals. Once the RU has been selected and the TT moves into the Planning stage, a TT PM will be assigned. In many TTs, the RU PM will serve as TT PM for the remainder of the project.

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Table 5.1-1 Initiation Stage Activities and Deliverables

STAGE	Business	TT PM	Process	Analytical	Engineering	Manufacturing	Regulatory	Quality	Sourcing/Supply Chain
Initiation	***Activities***								
	Develop Business Case	Coordinate Subject Matter Expert (SME) Assessments	Determine Preliminary Process Flow	Perform Initial Method Evaluation	Perform Permit Reviews (Licenses, Construction)	Perform Permit Reviews (Environment, Health, and Safety (EHS))	Evaluate Market Requirements	Perform RU Quality Assessment(s)	Perform RU Capacity Analysis
	Identify Technology Transfer (TT) Project Manager	Perform Initial Risk Assessment	Define Batch Size	Evaluate Laboratory Capabilities	Evaluate SU and RU Site Capabilities	Assess EHS Requirements	Develop Regulatory Strategy		Evaluate Import/ Export Requirements
	Perform Patent Review	Perform Due Diligence	Develop/Obtain Sending Unit (SU) Qualtiy Target Prodcut Profile (QTPP) and Product Specifications		Estimate Capital Requirements				Evaluate Material Sourcing
	Develop Product Strategy	Drive Receiving Unit (RU) Selection Decision	Evaluate Manufacturing Process Equipment Requirements						Determine Requirements for Tightly Regulated Materials
	Obtain Non-Disclosure and/or Confidentiality Agreements	Develop/Update Project Plan	Identify Scale-Up Needs						
	Deliverables								
	Business Case	Initial TT Charter	Process Flow/ Description	Laboratory Evaluation Report	Facility Fit Assessment	EHS Viability Report	Regulatory Strategy	Quality Assessment(s)	Preliminary Product Supply Chain Strategy, Including Map
	Product Strategy	High-Level Project Plan	Process Fit/Gap Assessment	Methods Evaluation Report	Permit Requirements	Permit Requirements	Regulatory Approval Timing		Critical Material List
	Patent Review Report	RU Due Diligence Report	Product QTPP		Capital Estimate				
	Non-Disclosure and/ or Confidentiality Agreements	Risk Assessment Report							
	RU Site Selection								
	Steering Committee Formed								

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The Initiation Stage PM coordinates an initial high-level risk assessment using cross-functional expertise and drives site selection by ensuring the completion of proper due diligence of the RUs under consideration. These assessments must be documented to support both the TT decision and site selection. One methodology for these assessments is risk-ranking and filtering.

One key deliverable the Initiation Stage PM must fulfill during the Initiation stage is the establishment of a charter with the steering committee. While the initial charter at this stage may not yet include all the details, it should document the boundaries, major risks, and assumptions used to determine whether the TT will proceed as a part of the product strategy, and it should specify the RU site that is selected. Boundaries and/or assumptions that should be considered during the Initiation stage to determine the scope of the TT include:

- Required time frame for completion
- Regulatory strategy options
- Estimated capital required
- Resource needs and constraints
- Scope of process to be transferred
- Target markets
- Annual production volume
- Site selection decision and criteria used

Finally, the PM creates an initial, high-level project plan estimating timing, resource needs, and budget.

5.1.3 Process

The Process function first determines the subject matter expertise required on the TT team for the product under evaluation by modality or stage of manufacture. For instance, a small-molecule drug substance would need a process expert who has worked with organic chemistry processing, while a filled-vial drug product would need an expert in sterile manufacturing. Second, the MAH Process function develops a preliminary process flow and works with the SU to develop or obtain the QTPP and/or specification for the product. Third, using the assumptions provided by the Business function, Process develops batch size options and assesses scale-up or scale-down requirements against the current process at the SU. Finally, working with the Engineering, Manufacturing, and Analytical functions, Process assesses the process fit and evaluates the facilities, equipment, and operations group of the potential RU site to determine its capability to manufacture the product, noting gaps and investments required.

5.1.4 Analytical

The Analytical function uses the SU's QTPP and/or specification to determine product testing requirements and to evaluate potential testing of raw materials, single-use and other components, and packaging. Test methods are categorized either as compendial or non-compendial for the target markets to define the complexity of method transfer from the SU. With this information, the Analytical function assesses the potential RU's laboratory capabilities and capacity and provides a high-level estimate of what would be required for the TT. This evaluation covers instrumentation needs, glassware, reference standards, personnel, external laboratory services, and timing. Evaluation of site analytical capabilities of potential RUs is an essential element of due diligence. At this stage, the Analytical function documents the laboratory and method evaluations.

5.1.5 Engineering

Evaluation of site manufacturing capabilities is another essential element of due diligence in selecting the most appropriate RU and deciding to proceed with the TT. This evaluation, also referred to as a facility fit assessment, includes such items as site utilities and infrastructure, processing equipment, and automation capabilities. The Engineering function, working closely with Process, Analytical, and Manufacturing functions, performs the facility fit assessment to form an initial rough estimate of capital that would be required. The case study in **Section 8.6** provides more detail on the facility fit assessment.

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5.1.6 Manufacturing

To perform site selection, the Business function needs to understand the potential RU's ability to implement the process. Capability extends beyond capital requirements. The Manufacturing function assesses the potential RU's resource flexibility, operational training program, previous and current manufacturing activities in the facility, organizational structure, current operating shifts, and process data management systems.

Environment, health, and safety (EHS) issues at the potential RU should be evaluated as well. An employee health and safety viability report can be used in evaluating potential issues. The Manufacturing function should assess the environmental and occupational permits or certifications that will be needed, along with the RU's ability to obtain them and their potential impact on the TT timeline.

5.1.7 Regulatory

Involving the MAH Regulatory function as soon as possible is critical to a TT project. A regulatory strategy should be developed to align the product strategy with the Business function and to establish boundaries and assumptions for the TT charter. Key elements of the initial regulatory strategy are filing category, regulatory assumptions, likely data requirements and source documents required, potential need for a health authority meeting, projected timing of regulatory submission to each health authority, and regulations applicable to the product in the region(s) in which it is to be manufactured and marketed.

The regulatory strategy is a living document that will evolve throughout the project. It should be reviewed at each stage for required updates and further refinement based on the knowledge gained at that stage. Therefore, the ultimate regulatory strategy may look significantly different than this initial draft as knowledge is transferred and more information becomes available.

The regulatory strategy typically would address the following:

- List of markets and the planned timing of submission for each market
- Proposed filing category for the TT and the justification for this determination
- Specific regulations applicable to the product and region(s) where it will be marketed
- Specific data requirements and source documents to support the change related to the TT that the governing regulatory agency would expect (e.g., sterility and aseptic processing, stability, qualification and validation considerations, clinical/bioequivalence requirements, and potential required permits)
- Key regulatory assumptions and risks that could impact the overall TT success and possible mitigation strategies
- Documents the RU will need to prepare for authoring, review, approval, and submission of the regulatory filing
- Product approval timeline
- GMP inspection history (lack of)
- Current GMP certificates

The Regulatory function also will perform due diligence of potential RUs for outstanding actions taken by regulatory authorities that would make the RU unsuitable for use.

5.1.8 Quality

The Quality function at the MAH performs the quality assessment as part of due diligence on the potential RUs. In preparing the quality assessment for each RU, which includes the RU site assessment, the Quality function typically asks the potential RU to complete a questionnaire and conducts a search for compliance history records. For external RUs, the compliance history records may be public, while the assessment of an internal site can involve records internal to the company. The purpose of the quality due diligence is to determine the maturity of the RU quality system, the RU's organizational mindset regarding GMPs, and its adherence to applicable regulatory requirements. Quality will provide a compliance risk evaluation of the potential RUs.

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5.1.9 Sourcing/Supply Chain

The Sourcing/Supply Chain function develops initial capacity-planning models using estimated yield and cycle times. These models are used to evaluate potential RUs for batch size, manufacturing cycle time, and on-site storage needs as part of the facility fit assessment. The capacity model should incorporate product volumes for development and validation activities as well as commercial scale. The Sourcing/Supply Chain function evaluates sourcing of the required materials, consumables, and components to determine any potential long-lead items and specially regulated materials (e.g., materials subject to national quotas), and it works with the Regulatory or appropriate function to identify special import/export requirements where permits would be needed. The Sourcing/Supply Chain function also prepares a preliminary supply chain strategy to assess the impact of each potential RU on the product value chain.

5.1.10 Best Practices and Additional Suggestions

Table 5.1.10-1 details additional ideas for consideration as well as best practices for the Initiation stage.

Table 5.1.10-1 Best Practices and Additional Suggestions for the Initiation Stage

Topic	Details
Establish a steering committee	• Establish the steering committee early to ensure that the steering committee, site leadership (if applicable), and executive leadership understand the project scope and requirements and that the steering committee is ready to respond as the primary escalation channel early in the project
Develop project charter	• At this stage, the project charter may be a simplified document summarizing the project's key requirements and scope that can be further defined as the project progresses
Document agreements and contracts	• Ensure a documented plan is in place that specifies who is responsible and accountable for specific activities to avoid assumptions, missed requirements, or misalignment so that the next stage (Planning) can start efficiently • Describe strategies in the charter that represent the full scope of the project and avoid documenting only short-term needs • Some aspects of the charter may be included in contractual agreements for clarity • Draft an initial regulatory strategy, including the expected approval timeline for the submission
Involve key functions early	• Involve key functions and personnel early during the Initiation stage to review the project requirements. This can help avoid missed requirements or changes to scope later in the project • Leverage expertise and lessons learned, if possible, from personnel with experience transferring products and with experience in the existing process
Ensure adequate talent, expertise, and focus on TT team	• Ensure the TT Team includes all key functions and expertise pertinent to the project and that team members have appropriate experience with transferring a product • Ensure personnel on the TT Team are either dedicated to the project or are held strictly accountable for their deliverables to prevent inattention to key details at critical times • Develop a backup plan to mitigate the potential loss of key personnel
Utilize risk management	• Use risk management tools to assess and document risks related to the process requirements, facility selection (e.g., gap assessment with process requirements), safety assessment, and quality requirements • Document formally the assessment of project and company requirements for the product being transferred, record any risks identified, standardize the risk classification of findings, and describe mitigation strategies or actions required. Include formal alignment of the steering committee and/or TT Team
Identify and plan for long-lead items	• Consider use of process analytical technology (PAT) as a component of the process control strategy and implement PAT as early in development as possible for new products • Identify long-lead equipment • Ensure critical path workstreams have a clear plan for the start and completion of activities • Identify and develop a plan for filling key project leadership positions to ensure personnel are in place and trained in time

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Develop checklists	• Prepare a checklist to assist in leveraging lessons learned from previous TTs • Use checklists for (1) requirements pertaining to capability, equipment, material, quality, safety, or personnel, and (2) documentation of procedural, physical, automation, and process requirements • A checklist may define documents or information to collect from facilities, teams, or third parties • Another checklist may define included documents to share across SU and RU to compare needs as the TT project launches
Arrange adequate funding	• Ensure that the project budget takes into account contingencies, such as changes to the scope of the project or the addition of any equipment, activity, resource, or requirement that may have been missed in planning. Revisit the budget assumptions regularly.
Consider other concurrent and near-future projects in planning	• Check how other projects might impact the project timeline and plan accordingly, paying special attention to shared work centers (e.g., Analytical, Regulatory) • Forecast resource capacity requirements and confirm timeline expectations as early as possible, covering the full scope of the project • Verify internal and external (as applicable) commitment to the project and identify any risks with resources and timing • Ensure that all involved parties are transparent in their needs for the key milestones, drivers, and ability to deliver

5.2 Planning Stage

The Planning stage begins after the site selection decision has been made and the initial charter has been approved; it is the start of the knowledge transfer from the SU. At this point, the TT PM is assigned, and the core TT Team at the RU is resourced. The members of the TT Team at the SU or MAH may or may not be the same personnel who performed the initial assessments in the Initiation stage.

Building on the risk assessments developed in the Initiation stage, the TT Team collaborates with the SU and/or MAH to perform detailed risk assessments to define all CMC and operational readiness requirements for the TT. This risk assessment becomes a living document that results in a revised and more detailed version of the project plan, which covers project timing, capital required, resource needs, and critical path items. Other key outputs of the Planning stage are the final TT charter and a TT agreement between the SU, RU, and MAH (see **Section 5.2.2**). Once the Planning stage activities and deliverables have been completed, the TT team should hold a stage gate meeting with the steering committee to obtain agreement for moving to the Knowledge Transfer stage. At this stage gate, the steering committee will approve the final charter and the project plan, which signals acceptance of the risks defined for the TT. **Table 5.2-1,** excerpted from the Reference Guide (**Figure 5.0-1),** shows the activities and deliverables for the Planning stage.

5.2.1 Business

The MAH Business function coordinates with the Sourcing/Supply Chain function to execute supply agreements (e.g., Master Service Agreement) with the RU and external laboratories. Critical to the outcome of the TT, supply agreements should be reviewed by the TT PM to ensure they contain the same boundaries and assumptions as the TT charter, align with the RU roles and responsibilities in the TT agreement, and support the requirements of the quality agreement. Many RUs ask that the supply agreements guarantee a fixed number of batches for both the initial qualification and the annual commercial production, as this allows for better resource planning.

Jointly, the Business functions and steering committee provide oversight of the project by approving resource and capital needs. A resource plan should be developed to understand staffing requirements both for the TT and for the long-term commercial needs, based on the following key inputs:

- Personnel required and organization of shifts to be used
- New equipment to be qualified and degree of automated vs. manual operational requirements
- Type of expertise needed
- Complexity of the technical requirements, including risk mitigation requirements

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Table 5.2-1 Activities and Deliverables for the Planning Stage

STAGE	Business	TT PM	Process	Analytical	Engineering	Manufacturing	Regulatory	Quality	Sourcing/Supply Chain
Planning	***Activities***								
	Approve Resource Needs	Define Resource Requirements	Detailed Process Risk Assessment	Detailed Analytical Risk Assessment	Define RU Requirements	Operational Requirements Analysis	Obtain Current Dossiers	Perform RU Quality Audit	Production Planning
	Approve Capital Plan	Define Critical Path Items	Align QTPP and Product Specification with RU	Define Analytical Equipment Requirements	Develop Capital Plan	Perform EHS Assessment	Interact with RU Regulatory Affairs Function	Define SU/ RU/Marketing Authorization Holder (MAH) Quality Management System (QMS) Responsibilities	Determine Product Logistics
	Interact With Customers	Develop Detailed Project Plan	Define Material Handling Requirements	Determine External Lab Needs	Write Capital Appropriation Request (CAR)	Initiate Permits	Interact with Health Authorities if Necessary	Define Documentation Requirements	Source Materials/ Disposables/ Packaging/ Consumables
	Sponsoring Steering Committee	Create Communication Plan	Define Commercial Process at RU	Define Sampling Plan	Write Change Controls			Define Comparability Requirements	Order Long-Lead Materials
		Implement Documentation Sharing	Define Bill of Materials					Develop Stability Plan	
		Hold Steering Committee Meetings							
	Deliverables								
	RU Supply Agreement	Final TT Charter	TT Change Control	Analytical Risk Assessment Report	Capital Project Plan	Operational Requirements Report	Current Product Filings and Annual Product Reports	RU Audit Report	Product Supply Chain Strategy
	External Lab Supply Agreement	Project Plan	Process Risk Assessment Report	Sampling Plan	Equipment/Facility Change Controls	EHS Assessment Report	Current Compendial Specifications	Quality Agreements SU/ RU/MAH/External Labs	Plant Schedule Availability
	Resource Plan	TT Agreement between SU/RU	Commercial Process Established		CAR Approval		Comparability Protocol	TT QMS	Long Lead Material Plan
	Stage Gate Approval	Critical Paths Identfied	Initial Bill of Materials		Change Controls			Quality Improvement Plan	
		RACI	Comparability Plan					Stability Plan	
		Communication Plan							
		Final TT Team Formed							
		Kickoff Event							

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Section 8.3 provides a case study of the importance of incorporating the learnings from previous projects into the resource plan during the Planning stage.

The Business function should provide input on the regulatory strategy, including volume requirements, required timing for market release, and the salability of the validation batches.

5.2.2 Technology Transfer Project Manager

Once the Business function determines that a TT will take place and has reached agreement with the RU, the TT PM is assigned. The TT PM, provides overall leadership and accountability for the remainder of the project. Therefore, the same qualities are important: knowledge across all relevant functional areas, ability to communicate across organizational levels, and skill in facilitation, negotiation, decision-making, and motivation. **Section 4.1** discusses the selection and role of the TT PM.

The TT PM should put in place a team representing the relevant cross-functional areas to begin planning and assessing the requirements for knowledge transfer. This team is established in collaboration with each site's functional managers. The case study in **Section 8.2** discusses the importance of establishing a cross-functional project team.

The TT PM works with the SU and RU to define the roles and responsibilities for all knowledge transfer requirements. If both the SU and RU are external, then the MAH should be included as a third party. A responsibility assignment matrix denoting those to be responsible, accountable, consulted, and informed (RACI) can prove valuable in documenting key responsibilities to ensure alignment and clarity throughout the project. Once negotiated, these roles and responsibilities should be documented in a TT agreement, or in a shared procedure for internal-internal TTs, and personnel can be assigned. The TT agreement can aid significantly in defining resource needs and in facilitating communication.

The TT PM plays the lead role during the Planning stage by coordinating the final risk assessments and developing the project plan. Using the detailed risk assessments, the TT PM will work with the TT Team to ensure that the project plan includes the schedule, critical path activities, and resource requirements. The TT PM also will ensure that the project requirements and timeline meet the needs of the business and stakeholders, taking into account the capabilities of the functions and systems supporting the project. The project plan typically covers:

- Capital procurement, installation, and qualification timing
- Procurement and release of all materials and consumables
- Process development and/or demonstration
- Analytical method validations, transfers, or qualifications
- Operational readiness requirements
- Plant-scale process validation
- Key milestones and critical path activities

Once the project plan has been developed, the TT charter can be expanded to include the communication plan. A communication plan and a system for sharing documents between the parties is necessary to facilitate knowledge transfer. The communication plan should define modes of escalation to ensure the TT Team can resolve any roadblocks that may occur.

At this time, the TT PM should hold a kick-off meeting with the final cross-functional TT team members (including both the SU and RU) and project sponsor.

5.2.3 Process

During the Planning stage, the Process function begins to specify the key elements of the commercial manufacturing process for the TT, including batch scale, unit operations needed, cycle time, and material handling requirements, which will result in an initial bill of materials. These knowledge elements are used

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by the Sourcing/Supply Chain function for production planning and plant logistics and by the Engineering function to design and order the required capital. The Process function aligns QTPP and/or specifications with the RU. Often, Process will develop a comparability plan to identify the information necessary to show that the product made at the RU is comparable to that of the SU. The Process function also performs a more detailed risk assessment to define the changes that will be required at the RU based on its equipment, scale, and capabilities. The risk assessment should include evaluation of the properties and requirements of the product being transferred. This risk assessment not only defines the knowledge transfer requirements from the SU, but also is used by Engineering to estimate the capital requirements. If the TT involves a new product or technology, the process needs to be designed such that it will run reproducibly at the RU to consistently meet the requirements of the QTPP and/or specifications.

The Process function, working with Quality and Regulatory, will define strategies for both process and cleaning validation at the Planning stage and ensure that the SU and RU align their change control processes. The Process function will issue the change request for the TT in the change management system as well.

5.2.4 Analytical

The Analytical function will perform a more detailed risk assessment to determine final analytical equipment requirements that will help determine analytical testing sites. Testing sites may be located at the SU, RU, MAH, or an external laboratory. The analytical risk assessment needs to include raw materials, packaging, in-process controls, and product, stability, and cleaning test requirements, specifically noting which methods are compendial. Adequate quantities of source materials, reference standards, and critical analytical reagents must be estimated and made available for analytical method validation, transfer, and future manufacturing. As the amount of sample source material can impact yield from validation batches, Analytical will work with the Process and Regulatory functions to develop an initial sampling plan that covers the number of samples taken, amount of sample required, tests performed, sample location, and whether the sample should be assessed only during process validation. This initial assessment of the sampling plan allows for planning of adequate analytical resources during qualification.

5.2.5 Engineering

Working closely with the Process function, Engineering will develop the capital requirements for the chosen RU, which are then documented in a capital appropriation request (CAR). Even before the TT user requirements specifications (URS) or CAR are complete, it may be necessary to decide to order long-lead equipment to meet TT timing requirements. Engineering also should be involved in change control for facility retrofits as well as laboratory modifications.

If the RU is a CMO or CDMO, the responsibilities of the Engineering function may reside at the contract site with strong oversight from the TT Team. The TT PM works closely with Engineering to coordinate the timing for equipment and facility installation, qualification, and training to align with the overall project plan.

5.2.6 Manufacturing

In the Planning stage, Manufacturing coordinates with the TT PM to ensure that the project timeline considers all necessary steps for full operational readiness of the RU site. Operational requirements of the technology being transferred should be considered relative to the staff capabilities at the RU, permit requirements, and operational training that will be needed for the new process and modified facility. Manufacturing also will work with the Engineering and Process units on plant design to define URS and CAR requirements.

EHS considerations must be assessed against the RU's procedures and experience so necessary permits can be prepared. The EHS assessment includes protective equipment, product biosafety, aseptic practices, hazardous materials, and potential vent and waste streams. A report on the operational requirements and an EHS assessment report will focus attention on the areas that need to be addressed in the project plan.

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5.2.7 Regulatory

The Regulatory function will further define the regulatory strategy in the Planning stage, identifying any items that could impact the Business function's product strategy (see **Section 5.1.1**). The regulatory strategy should consider the documents that will be needed and include a projected timeline for key milestones such as drafting, review, and approval. These will later be expanded into a detailed submission tracker.

The regulatory strategy should include the regulatory risks, with mitigation options, and the regulatory assumptions identified by both the MAH and the RU. Collaboration between the MAH and RU Regulatory and Quality functions at this stage is important in understanding the risk tolerance and concerns of each, both in general and specific to the product. Establishing clear understandings at this stage can help avoid delays and frustration later in the process.

The Regulatory function communicates the submission requirements to the Process, Analytical, Engineering, and Manufacturing functions, which will perform detailed risk assessments. Based on the level of risk the MAH is comfortable with, Regulatory may be able to define opportunities to reduce complexity and accelerate regulatory submission and approval. Sources of risk that often are considered in the regulatory strategy include:

- Stability strategy
- Analytical method transfers
- Number and timing of validation batches
- Process changes or improvements
- Amount of data and activities the RU can leverage from the SU

The MAH Regulatory function may meet with regulatory authorities to review the proposed regulatory strategy to ensure success.

The RU Regulatory function works with the MAH Regulatory unit to obtain current CMC information from the registered dossiers and to evaluate any compendial or compliance standards that have changed. It also is useful to obtain information about any changes implemented across the product's lifecycle to better understand challenges in manufacturing or testing of the product. Additionally, the outputs in the submission (e.g., CQAs and critical process parameters (CPPs)) should be provided to the RU so the RU can understand any critical challenges with the product and determine the inputs necessary to meet those requirements. If the TT is for a new product, the MAH Regulatory function will define the requirements based on the target markets and manufacturing locations.

5.2.8 Quality

The MAH Quality function will work with the RU Process function to document in the QTPP and product specification the requirements that will ensure the quality, safety, and efficacy of the product. In this stage, quality agreements must be developed concurrently and executed, as required, between the MAH and RU, and between the RU or MAH and additional GMP providers and external laboratories.

Based on the target market(s), current stability requirements, and commitments made to regulatory authorities, Quality will work with the Process, Analytical, and Regulatory functions to prepare the stability requirements. Regulatory uses the stability plan to further develop the overall regulatory timeline to ensure that all conditions of the current commitments can be met.

Using the regulatory strategy, Quality also begins GMP audit-readiness activities at the RU to ensure full inspection readiness when needed in the Execution stage. First, the MAH Quality function performs an audit of the chosen RU so that all compliance gaps impacting the TT can be addressed prior to physical implementation of the cross-functional requirements. When significant quality changes are required, a quality improvement plan will be developed; however, the quality assessment performed during the Initiation stage should have identified if a site has a significant compliance gap.

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Quality also performs a gap assessment of the SU, RU, and MAH quality systems and negotiates an aligned QMS for the TT. This alignment should be performed even when the SU and RU are both internal sites, as interpretation of the global QMS may differ across regions.

5.2.9 Sourcing/Supply Chain

Once the decision to perform the TT is made, production planning begins. The Sourcing/Supply Chain function works closely with the TT PM to evaluate the time that will be needed for the TT, based on the time needed for Engineering installation and qualification, the number of engineering batches to be performed, the number of qualification batches needed, and the time needed for stability studies and regulatory approvals. Planned downtime in production, such as for maintenance (reactive or preventive) and cleaning, also should be considered.

During the Planning stage, the target market volume also is used to determine future RU capacity and resource needs for commercial supply. These requirements typically are part of the supply agreement negotiated with the RU during the Planning stage.

During this stage, the sourcing of raw materials, consumables, and packaging components should begin with identification of sources for the items with the longest lead times and sole-sourced items. Any changes to the supply of critical materials identified as part of the TT risk assessment also should be addressed.

The Sourcing/Supply Chain function also evaluates overall product logistics to further develop the product supply chain strategy. At this stage, the product supply chain strategy should show import/export considerations, distribution, and interim points involved with both the product and materials. A detailed supply chain map showing distributors and manufacturers for key components is a key output to identify supply chain risks, consider second sources of critical materials, and facilitate future troubleshooting.

5.2.10 Best Practices and Additional Suggestions

Table 5.2.10-1 describes additional ideas for consideration and best practices for the Planning stage.

Table 5.2.10-1 Best Practices and Additional Suggestions for the Planning Stage

Topic	Details
Hold a well-organized project kick-off	• Kick-off meeting should be well organized to cover all relevant activities and functions • Meet face to face with the TT Team if possible. In the absence of a face-to-face meeting, allow adequate time (e.g., 1-2 days or half-days) for discussion and alignment around the project plan and key milestones • Establish an online repository for document-sharing
Understand established conditions	• Identify and share with the RU the existing manufacturing process conditions at the SU, including material, people, and process flows • Determine and communicate the changes that may need to be made at the RU
Define prerequisites for introduction of product to new site	• Ensure that the proposed process, development data, and control strategy meet all expectations and requirements before proceeding • Ensure requirements for the RU and SU are well understood for all processes, analytical methods, documents, and forms required to initiate activities related to the project start, e.g., change control, material setup • Ensure key timelines/actions necessary to meet each requirement are understood
Perform risk assessments	• Conduct or update detailed process and analytical risk assessments so the TT Team will be well informed about the primary areas of risk in this transfer, allowing them to create risk mitigation strategies
Ensure project oversight and team engagement	• Assign one person from both the RU and SU as team leads/PMs to be accountable for the project throughout its lifecycle and to be ultimately accountable to the steering committee • Ensure that team members are participating, engaged, following through with their assigned actions, and meeting timelines

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(Table 5.2.10-1 Continued)

Topic	Details
Engage vendors early	• Engage all vendors (e.g., equipment, excipients) early to identify tentative lead times and begin drafting a project timeline • Ensure that the RU understands the quality requirements for raw materials and drug substance to ensure that selected vendors will meet the quality requirements outlined in the marketing authorization • Stay in contact with vendors throughout the process, as it may take months to receive the project award and approval, delaying placement of orders • Identify backup suppliers as alternative sources
Engage Regulatory function early	• Further define the regulatory strategy with project assumptions and constraints and with regulatory requirements • Discuss the regulatory strategy with the TT Team to ensure alignment on assumptions and the basis for the strategy • Engage the TT Team to define variations that might cause a deviation to the regulatory strategy or might reduce complexity • Consider the need to proactively engage a regulatory agency if required to support the strategy and confirm needs. Meetings may be advisable if a unique situation exists or if the TT could create a situation that may impact supply (e.g., drug shortage)
Create timeline that ensures sufficient time for drafting and review of regulatory filing	• Ensure the project timeline includes key milestones related to drafting, review, and submission of filing and allows for regulatory agency approval • Ensure the regulatory filing strategy is realistic so the project timeline is accurate, and the Business function can plan • Develop a stability plan, making sure to include any areas where the product will be marketed • Coordinate activities and documents required for submission to avoid extensive loss of finished product due to shelf-life restrictions while awaiting submission approval • When developing the timeline, anticipate that additional questions may need to be addressed after initial submission, and that SMEs may need to be involved in developing responses
Documentation quality	• Understand that quality documentation is critical to ensure that the regulatory filing will be approved within the expected timeline, supporting support product launch, demand requirements, and business needs • Have Regulatory review during drafting any documents that are necessary for the regulatory filing • Consider conducting a risk review of documents that may be reviewed during an inspection that could impact the filing
Maintain communications	• Ensure that changes are communicated throughout the TT Team to allow all functions to adjust and adapt in a timely manner
Understand how small changes can impact the regulatory strategy	• Minimize changes that are not necessary to keep the regulatory strategy straightforward. Additional changes will require more bridges to be built to justify changes • Ensure that all changes during a commercial-commercial TT are vetted through Regulatory, as the cumulative effect of changes could result in a need for further regulatory action that may cause additional work and delays • Ask the MAH Regulatory function to ensure that regulatory documentation to be used for knowledge transfer reflects the current process, test methods, and specifications
Forecast 12-18 months for project and/ or commercial needs	• Forecast supply needs, including needs for both the TT and initial post-transfer manufacturing, as far in advance as possible (in the range of 12-18 months) • Ensure that the RU functions capture and understand the needs relating to timeline, lab testing, labor, and document approval to ensure that personnel can plan ahead and execute on time • Plan early for the project's long-term/commercial needs to avoid capacity restriction risks later
Plan appropriate engineering work during TT	• Exercise care when leveraging from similar processes • Plan for multiple batches to verify scale-up • Leverage expertise early (both in product and process and on equipment) to assess requirements and anticipated timelines • Transfer may require significant development work. Incorporate principles of quality by design and stages 1-3 of the process validation lifecycle to fully understand the process
Start reviewing equipment needs early	• Ensure equipment at the RU is equivalent to equipment at the SU, or that comparability is demonstrated • Ensure URS are well defined so that pricing is accurate, CAR approvals can follow quickly, and equipment requirements are understood upon award • Ensure that lead times from the vendor(s) are understood

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5.3 Knowledge Transfer Stage

The Knowledge Transfer (KT) stage significantly overlaps with the prior Planning stage and the following Readiness stage. During Planning, key pieces of the KT are needed to finalize the TT charter and perform risk assessments. KT often is still in progress while key activities in the Readiness stage are being developed. Throughout the TT, and especially in the KT stage, any events or changes that occur at the SU that impact the product being transferred require immediate communication to the TT PM. KT can be a highly iterative and challenging process. At the KT stage, all cross-functional roles involved in performing the TT are fully engaged.

The TT agreement, quality agreement, TT charter, and project plan that were completed in the prior stage allow the SU and RU to execute the full transfer of information, systems, data, technical requirements, and practical experience necessary to deliver the CMC and operational requirements to the RU. The TT Team members will work very closely during the KT stage, employing the communication plan and document-sharing platform developed during the Planning stage.

KT extends beyond the direct TT team members. All supporting personnel across the SU and RU need to understand the TT requirements, product and regulatory strategies, and success factors as defined in the TT charter so workloads can be prioritized across the organization and timelines will be met. In addition, supporting personnel across the SU and RU may hold key information and expertise based on experience with the intended equipment, a similar process, or a previous project. This additional expertise can assist in KT.

During the KT stage, a proactive knowledge management (KM) approach can improve KT, create efficiencies between the SU and RU, and facilitate more effective transfer of both explicit (documented) and tacit information held by personnel. The transfer of tacit information may offer the greatest opportunities for improvement, as this type of information often may not be uncovered unless a failure result leads to its discovery. A robust KM approach that includes review of TT performance also can provide more detail that can be used to refine the control strategy after the TT is complete and to improve future transfers. Every opportunity should be taken to leverage KM to preserve, transfer, and increase knowledge about the product and process ***(2)***.

The TT package developed in the KT stage will include all the details needed to reproducibly manufacture, test, and release the product being transferred. During the stage gate at the end of the KT stage, the steering committee ensures that the TT package is complete and assesses the risk factors, with mitigation strategies, that remain after the KT.

Table 5.3-1, excerpted from the Reference Guide (**Figure 5.0-1),** shows the activities and deliverables for the KT stage.

5.3.1 Business

With the required contracts and nondisclosure agreements in place, the Business function focuses on refining and executing the resource plan developed in the Planning stage. The Business function monitors progress of the interactions between the SU and RU and removes any contractual roadblocks that may limit KT activities. The case study in **Section 8.10** discusses best practices and the benefits of proactively driving KT between the SU and RU to help ensure a right-first-time project execution mindset.

5.3.2 Technology Transfer Project Manager

Using the TT agreement developed during the Planning stage, the TT PM, with the TT Team, facilitates KT across key functions in the SU and RU (and third parties, as applicable) to ensure timely and complete knowledge-sharing. The KT is typically documented by the SU in a TT package containing all the information needed to manufacture the product robustly.

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Table 5.3-1 Activities and Deliverables for Knowledge Transfer Stage

STAGE	Business	TT PM	Process	Analytical	Engineering	Manufacturing	Regulatory	Quality	Sourcing/Supply Chain
Knowledge Transfer	***Activities***								
	Removes Roadblocks	Communicate with SU	Define Critical Process Parameters	Evaluate Material and Product Method Requirements	Identify Plant/ Analytical Equipment Requirements	Assess SU Master Batch Record	Obtain SU Product Label	Perform Quality Risk Assessment	Materials/Disposables/ Packaging Supply Agreements
	Executes Resource Plan	Obtain Process Knowledge from SU	Determine In-Process Tests	Evaluate In-Process Test Methods	Identify Disposables	Define Personal Protective Equipment (PPE) Requirements	Refine Regulatory Strategy	Perform Material Supplier Qualifications	Obtain Material Specifications
	Configure Material Requirements Planning/Enterprise Resource Planning Systems for New Product Introduction	Facilitate Writing of RU Specifications	Finalize Process at RU	Evaluate Cleaning Methods	Write Capital Validation Master Plan	Define Operations Documentation	Provide Regulatory Support	Develop Training Curricula	Source Reference Stds
			Define Cleaning Procedure and Limits	Evaluate Microbial Methods	Design/Order Capital Items		Review Stability Plan	Establish Expiry/ Retest Date	Develop Bill of Materials
			Assess Rework/ Reprocess	Order and Qualify Reference Standards	Design Environmental Disposal				
			Write Validation Master Plans (VMPs)	Assess Compendial Methods	Write Change Controls				
			Define Process Monitoring Requirements	Define Harmonized Methods					
			Demonstrate Process						
	Deliverables								
	Stage Gate Approval	SU Technical Package	RU Technical Package	Reference Standards	User Requirements Specification (URS)	PPE Requirements	Updated Regulatory Strategy	Quality Risk Management (QRM) Report	SU Material/Packaging/ Consumables Specifications
	Change Orders	RU Material Specifications	Mass Balance	Analytical Method Validation Plan	Equipment Validation Master Plan		Health Authority Interaction	Supplier Qualification Reports	Bill of Materials
		RU Product Specification	Waste Characterization		Engineering Drawings		Stability Plan	Compliance Training	Qualified Supplier List
			Process VMP		Change Controls				
			Cleaning VMP						
			Lab Demonstration Runs						

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The TT PM should be sure to include the sharing of practical experience, or "know-how," which may or may not be documented. To capture tacit knowledge, the TT PM should coordinate on-site interactions between SU and RU personnel, allowing the SU to demonstrate activities specific to the product or process being transferred. This knowledge also should be captured in the KM system being used for the product.

KT is key to a successful handover from the project team. During KT, the TT team should both leverage learnings from previous projects and capture lessons that are learned as part of the current TT. The case study in **Section 8.5** illustrates the importance of KT and of leveraging past learnings.

The TT PM establishes project controls, including lifecycle document management, and now actively manages the project scope, timing, budget, and risk factors according to the TT charter requirements. Regular TT Team meetings are used not only for KT but also for timely execution of the deliverables, identification of roadblocks, and tracking of additional action items that may be specific to the ongoing TT.

5.3.3 Process

If the product is already established commercially at the SU, the SU shares process performance qualification (PPQ) information for the process with the RU Process function. This information will include CPPs and their impact on CQAs, in-process tests, and proven acceptable ranges for the manufacturing process. The SU should share learnings from the implementation of commercial or pilot-scale manufacturing processes, along with process improvements that have been made. To ensure full understanding of the process history, lessons learned, challenges, and early development of the process, the SU and/or MAH also should share with the RU historical performance and development data wherever possible. For new products, the RU will work with the SU to define the PPQ requirements and ensure their feasibility at the RU facility. These PPQ requirements will include a list of CPPs that are derived from quality risk assessments and laboratory and or pilot-scale data. For more information on parameter classification and the quality risk assessment process, please see PDA *Technical Report No. 60*: *Process Validation, A Lifecycle Approach* ***(5)***. Using the PPQ requirements, the Process function will determine the data to be collected and will define how this information will be captured to ensure data integrity.

In this stage, the Process function writes the process and cleaning validation master plans (VMP) as key deliverables, using the strategies developed during the Planning stage and the RU's standard operating procedures (SOPs). The SU shares with the RU all information required for cleaning validation, including solubility, toxicology category (with data, if possible), and safety data sheets, so that the cleaning VMP can document the cleaning limits.

The process VMP outlines the PPQ requirements covering:

- Process description, including control strategy and CPPs
- Batch size and expected yield
- Raw material qualification plan
- Number of PPQ production runs, detailing any matrix or bracketing assumptions
- Hold-time strategy
- Rework or reprocess steps allowed
- Stability strategy
- Success criteria

Using the risk assessment from the Planning stage, the Process function performs activities needed to define the control strategy, including CPPs that impact CQAs and their allowed ranges, in-process testing, and control systems. For drug products, these activities may include, for example, mixing, hold time, filter integrity, container closure, and transport or shipping studies. When possible, the RU should demonstrate the process in the laboratory to ensure full understanding of the knowledge being transferred from the SU.

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5.3.4 Analytical

In the KT stage, the Analytical function at the RU develops the full method-implementation strategy, known as a validation plan, using the sampling plan, actual non-compendial methods used at the SU, and compendial methods for the markets included in the business and regulatory strategies. The analytical validation plan must align with the QTPP and/or specification requirements provided by the SU or documented in the marketing authorization file. When a marketed product is being transferred to a different site, validation may be performed through the analytical method transfer plan. The Analytical function determines the requirements for reference standards to allow for the ordering of reference standards for method qualification and testing. During the KT stage, potential opportunities for harmonization of analytical methods are identified and an implementation plan for use of these methods is discussed with the Regulatory function. The Analytical function documents the agreed-upon approach to implementation of all analytical TT requirements, including the use of any contract testing sites.

5.3.5 Engineering

The Engineering function performs a detailed facility-fit assessment in the KT stage. In doing so, the Engineering function uses the knowledge being transferred from the SU to compare the existing (for approved product) or proposed manufacturing process with the RU's capabilities. Process steps can be modified as part of the final facility fit to facilitate execution of the manufacturing process at the RU. However, any process modification should be proven to work prior to the Readiness stage. Engineering also should design and implement an appropriate waste management system covering all steps from waste identification to disposal. The Engineering function writes the change controls for the facility, often as part of the overall TT change control.

Working closely with the Process, Analytical, Manufacturing, and Quality functions, Engineering will define all facility, equipment, automation, and laboratory requirements and document these in the URS. Single-use components (sometimes known as disposables) also should be part of the URS development. A layout should be used to confirm single-use component design and to ensure all qualification requirements (i.e., extractables/leachables) are identified. The key aspects of the equipment involved in the TT, defined in the URS, are used in designing and ordering resources. In the KT stage, vendors are identified, engineering drawings are completed, and all required capital equipment is ordered. The timeline for Engineering function activities and deliverables can be highly influenced by how the URS is defined, which vendors are selected, whether on-site vendor support is used, and how well standard vendor activities can be leveraged for the capital qualification plan. For instance, some vendors offer validation packages or will participate in equipment qualification to ensure a smooth process. Use of the factory acceptance test (FAT) or site acceptance test (SAT) as part of the equipment qualification may also reduce effort and documentation requirements during the Readiness phase.

Engineering works with the Quality and Regulatory functions to develop a capital VMP that describes the strategy for qualification or requalification of the facility. The capital VMP also may include the minimum requirements for developmental purposes in order to execute an engineering batch.

5.3.6 Manufacturing

The RU Manufacturing function works with the Engineering and Process functions to ensure that the RU plant has adequate capability to complete the TT. Manufacturing will review the SU's master batch record, if applicable, and define the operational documentation requirements for the TT (e.g., batch records, SOPs, work instructions, training). Throughout the KT stage, all cross-functional areas should be interacting with Manufacturing to ensure plant design and training result in a process that can be operated in a reproducible manner to make a product of consistent quality. SU personnel or experts should meet with RU personnel to convey practical experience in making and/or developing the product, including lessons learned from process deviations, if applicable.

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In the KT stage, Manufacturing should define personal protective equipment requirements to protect both operator and product so that the Sourcing/Supply Chain function can order the needed equipment.

5.3.7 Regulatory

During the KT stage, the Regulatory function supports the RU's ability to receive the process, refines the regulatory strategy, aligns with validation approaches, continues planning activities, and focuses on building key relationships. KT from the SU or MAH Regulatory unit should include relevant health authority interactions relating to the regulatory process, historical audits, and licensure. Regulatory should communicate project-specific risks to the TT Team to ensure alignment and understanding.

The regulatory strategy developed during the Initiation and Planning stages should be further developed during the KT stage, considering the finalized stability plan. Regulatory also should consider the need for any changes to the primary and secondary labeling.

The revised regulatory strategy also should address the timing and content of planned interactions with regulatory authorities, including meetings, scientific advice, protocols, and/or correspondence. Important considerations for the regulatory strategy include definition of the matrixing approaches for qualification and validation as well as any approach to be taken to prove comparability. The MAH may consider submitting correspondence or requesting a meeting with the regulatory authority if the acceptability of the comparability strategy is uncertain. The MAH also should consider submitting a post-approval change management protocol to gain concurrence on the approach being used for this TT or to potentially reduce the reporting category for future submissions addressed in the protocol.

5.3.8 Quality

At the KT stage, the RU Quality function performs a quality risk assessment to evaluate the ability of the RU's quality systems to make a safe and efficacious product. In conducting this risk assessment, the Quality function will use the knowledge and information that has been shared by the SU. The quality risk assessment will rely on findings from audits, outcomes from the quality improvement plan (if applicable), SU quality metrics, and market actions. For a product already being manufactured under a marketing authorization, the quality risk assessment also can examine deviations and corrective actions at the SU as well as process trends, product complaint history, and annual product reports as available. The Quality function should perform compliance training and address any gaps identified by the quality risk assessment.

Quality also finalizes the stability plan based on the requirements in the QTPP and/or specifications and ensures its alignment with the regulatory strategy for the TT.

The RU Quality function works with the Sourcing/Supply Chain function to identify, schedule, and perform supplier qualification audits for all materials being sourced for the TT. These functions will document the findings and recommendations in a supplier qualification report. They also will define expiry/retest dating requirements for both materials and product.

Quality also plays a key role in assessing the compliance of the analytical validation plan, change controls, project plan, and process, cleaning, and capital VMPs.

5.3.9 Sourcing/Supply Chain

The RU Sourcing/Supply Chain function, with the Process function, defines the RU bill of materials using the material requirements that are shared during knowledge transfer. Key considerations include the SU bill of materials, storage requirements, shipping regulations, regulatory classification, and specifications. Lead time for material delivery is often a critical path item for the project plan. The

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RU should consider obtaining samples of materials from suppliers for testing and performing either laboratory runs or a future engineering batch. Storage and handling requirements may also drive capital investment, so the capacity plan developed in the Initiation stage should be reevaluated.

Capacity-planning also impacts the recruitment and training of personnel required to perform the TT. The supply chain map lays out the projected flows from materials-in to product-out, including the testing needed to support production. Therefore, it serves as a key input for resource-planning in the Operations and Quality Affairs/Quality Control units, and other critical functions.

5.3.10 Best Practices and Additional Suggestions

Table 5.3.10-1 describes best practices and presents some additional ideas for consideration in the KT stage.

5.4 Readiness Stage

In the Readiness stage, the TT Team has all documentation, equipment, and analytical methods in place, and the RU site is prepared to safely perform process qualification in a compliant manner. Preliminary KT has occurred, so the RU can finalize the control strategy, qualify the analytical methods, and complete the facility modifications. Activities in this stage often are performed in parallel with the KT stage, which requires an iterative process in document development and approval. Because the risk assessment is a living document, risk identification and risk mitigation strategies should be constantly reviewed throughout the TT project. A practical plant capability assessment should be conducted prior to PPQ. During this assessment, preliminary water or engineering runs may be performed as part of the overall project plan.

As the name implies, Readiness sets the stage for a successful PPQ at the RU. In the stage gate for Readiness, the steering committee determines whether all requirements are in place for batch execution and gives approval to proceed with the engineering batch. The case study in **Section 8.11** highlights the importance of executing a stage gate meeting to ensure that all Readiness requirements have been met prior to proceeding to the Execution stage.

Table 5.4-1, excerpted from the Reference Guide (**Figure 5.0-1),** shows the activities and deliverables for the Readiness stage.

5.4.1 Business

At this point of the TT, the RU Business function should be focused on completing all resourcing requirements so personnel can be trained in preparation for PPQ. The Business function also actively monitors the TT Team's ability to achieve the business objectives defined in the TT charter, including budget, launch date, and projected volume.

As the MAH Business function actively engages customers for the product being transferred, it must effectively communicate any changes to the business strategy when they occur. This allows the TT PM to assess the impact of the changes to the overall project assumptions and risk.

5.4.2 Technology Transfer Project Manager

During the Readiness stage, the TT PM monitors all workstreams and team members executing the requirements for plant readiness documented in the project plan. With many critical activities coming together, in order to proceed to the next stage the TT PM's thorough execution and coordination of the communication plan is critical in monitoring progress, challenges, and obstacles. The TT PM must continually evaluate the impact of changes on the TT charter, facilitate decision-making, and actively manage risk.

The TT PM also documents the PPQ requirements governing the TT in a TT protocol, summarizing the process and cleaning VMPs, the analytical validation plan, and the stability and sampling plans. If

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Table 5.3.10-1 Best Practices and Additional Suggestions for the Knowledge Transfer Stage

Topic	Details
Prepare package of materials for the RU site prior to KT kick-off	• Ensure a clear, well organized TT package is accessible to the RU team. Ensure the Regulatory function confirms the version of key documents to be provided • Ensure the RU has access to key documents and learnings from the SU to avoid reinventing existing documents or repeating past mistakes • Ensure data is in a transferable format and does not require manual reentry
Establish a communication plan for effective collaboration	• Avoid sending lengthy or complex documents or a very large number of documents to the RU • Arrange for the SU/MAH to provide training sessions to orient the RU and personnel to explain documents and to share expertise during the review process
Leverage expertise from the existing process and/or previous projects	• Review and/or incorporate knowledge from similar projects or processes to avoid additional work, repeating issues, etc. • Ensure that the SU provides key personnel (e.g., scientists, engineers, analytical personnel, production operators) to support and/or train the RU staff • Encourage face-to-face interaction and site visits to support governance and to facilitate engagement between RU and SU SMEs • Obtain samples of product for development • Capture tacit knowledge • Confirm sourcing of reference standards
Provide the RU with historical data and performance of product manufactured by existing processes	• The SU should provide the RU with historical data not already included in the TT package. For an existing commercial process, for instance, this may include several years of commercial data such as release testing, CPV reports, annual product reports (APR), deviations, and complaints • The RU may leverage such learnings from the SU as process control and operational performance, as well as employee training and batch record needs
Perform comparability assessment and define strategy	• Depending on project scope, the TT Team may perform a comparability assessment and review learnings from the SU and MAH to define overall project strategy (i.e., scaling up, creating a new process, or proving equivalency) • Develop a decision tree to consider whether process data is sufficiently equivalent between SU and RU • Plan for engineering batches or appropriate analyses to demonstrate comparability or equivalence
Use a quality-by-design approach	• A quality-by-design approach might be used to fully understand the process design space before starting engineering batches. This will depend on the scope of the project (e.g., transferring an existing product with commercial history and available development data or scaling up from development) and on the business strategy (i.e., whether development data and design space will be made available to the TT Team)
Leverage risk assessment to define KT requirements	• Prepare a clear, well-organized plan documenting the process requirements against all site, company, and industry requirements, identifying any gaps that will need to be addressed in the next stage • Use risk assessment throughout the TT project to identify any gaps in process, methods, materials, and people
Formulate an adequate process data-monitoring plan	• Prepare a plan for how process data will be recorded, by whom, and in what system • Consider data integrity requirements as well as any requirements for inclusion in a data lake or other enterprise repository • Determine if the RU requires additional software capabilities to match the SU's • Ensure appropriate data are collected to support project execution/completion and to meet requirements • Ensure data monitoring requirements are supported by the RU's information technology infrastructure
Plan adequately for validation needs or leverage existing knowledge	• Leverage learnings, such as those for cleaning validation, and plan for sufficient equipment to allow time for in-between cleanings while meeting process-throughput requirements • Identify and leverage SMEs with knowledge on existing or similar equipment systems at both the RU and the SU • Review validation information for existing or similar equipment systems, if available, for lessons learned and validation strategies • Leverage support from the vendor for new equipment and technology, which may include on-site visits during the next stage
Hire/train RU personnel	• Ensure that needed personnel are hired/trained at RU to avoid errors and repeat work throughout the TT

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Table 5.4-1 Activities and Deliverables for the Readiness Stage

STAGE	Business	TT PM	Process	Analytical	Engineering	Manufacturing	Regulatory	Quality	Sourcing/Supply Chain
Readiness	***Activities***								
	Completes Resource Plan	Write TT Protocol	Develop Control Strategy	Write Method Validation/ Transfer Protocols	Perform Factory Acceptance Test (FAT)/Site Acceptance Test (SAT)	Write Master Batch Records	Review Product Labeling Mock-Ups	Review and Approve Documentation	Order Materials
	Communicate Changes to Business Strategy	Remove Roadblocks	Demonstrate Process Monitoring	Execute Method Validations/Transfers	Install Capital Items	Write Standard Operating Procedures (SOPs) and Work Instructions	Review Change Order Notifications	Write Procedure for Batch Release	Define Distributors
	Establish Customers	Facilitate Decisions	Write Process Validation Protocol	Confirm Compendial Method Viability	Set Up Control System	Perform EHS Risk Assessment	Develop Submission Tracker	Train Quality Assurance Personnel on Process	Set Up Inventory Control Software Requirements
			Write Cleaning Validation Protocol	Write Stability Protocols	Write Capital Qualification Protocols	Perform Operations Training	Identify Necessary Permits & Licenses	Release Materials	
			Establish Hold Points	Perform Analytical Training	Perform Equipment/ Facilities Qualifications	Perform Pre-Cleaning		Audit Vendors and Suppliers	
			Develop Sampling Plan		Write Preventative Maintenance Plan	Set Up Warehouse		Develop Preapproval Inspection (PAI) Readiness Plan	
			Write Shipping Validation Protocols		Set Up Spare Parts Requirements	Execute Cleaning from Installation		Review Product Labeling Mock-Ups	
			Perform Process Training						
	Deliverables								
	Resources Available	TT Protocol	Control Strategy	Validated or Qualified Methods	Installed Equipment	Master Batch Record	Final Regulatory Strategy	Approved Documentation	Supply Chain Map
	Current Business Strategy		Process Validation Protocol	Qualified Reference Standards	Capital Qualification Protocols	SOPs and Work Instructions	Submission Tracker	Equipment Readiness Approval	Materials Delivery
	Customer Contracts		Cleaning Validation Protocol	Stability Protocols	Capital Qualification Reports	EHS Risk Register	Internal Product Labeling Approval	Deviation/Corrective Action/Preventive Action (CAPA) Closure	
	Stage Gate Approval		Hold Points	Trained Personnel	Preventative Maintenance Program	Trained Personnel		Qualified Vendors and Suppliers	
			Sampling Plan		Spare Parts List	Clean Equipment		PAI Readiness Plan	
			Shipping Validation Protocol					Internal Product Labeling Approval	

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multiple TTs are taking place across the supply chain for drug substance production, drug product manufacture, and packaging and labeling, the project plan may include details of the product value stream.

5.4.3 Process

The Process function in the Readiness stage focuses on having all data necessary to finalize the product control strategy. The control strategy defines the criteria to ensure a robust and repeatable process for making a product that meets the QTPP and includes hold points, in-process sampling, CPPs, process ranges, and critical unit operations. Close collaboration with Process, Engineering, Analytical, and Manufacturing is needed to ensure deployment of an effective control strategy.

In accordance with ICH Q10, the classification of process parameters and their ranges typically is determined by the results of laboratory batches and the increasing knowledge gained during the clinical phases of development. For commercial products, the RU can leverage a control strategy already in use by the SU by using its historical data and practical experience. For new products, process characterization studies at laboratory- or pilot-manufacturing-scale will provide understanding of the design space and the edges of failure for product quality and process efficiency. The results of such studies will be proven acceptable ranges and normal operating ranges for all unit operations. These lab runs also may include evaluation of product compatibility with single-use systems or assessment of equipment materials of construction proposed for the manufacturing process.

Final engineering batches that demonstrate the control strategy should be performed prior to PPQ. To complete the Readiness stage, the entire process should be tested to demonstrate operational readiness before moving into PPQ. Such readiness includes any final modifications, document or procedural readiness, and third-party resources. The TT Team should avoid introducing any new conditions or variables to the process once the Readiness stage has closed.

In the Readiness stage, Process writes the process validation protocols capturing all process parameters, the sampling plan, and the analytical testing strategy. The strategy for testing during PPQ typically includes additional sampling locations and requirements compared to routine commercial production. Shipping and cleaning validation protocols also are written and approved during this stage.

5.4.4 Analytical

The RU Analytical function executes the validation plan developed during the Planning stage. In Readiness, all method validation and/or transfer protocols covering raw materials, product, packaging, cleaning, microbiological testing, and in-process tests are written, approved, and executed. Compendial methods should be performed at the RU to verify suitability. The Analytical function also writes the stability protocols prior to the Qualification stage. New analytical equipment will need to be installed and qualified before the method validation activities begin, and analytical personnel will need to be trained.

In the Readiness stage, the Analytical function also supports the Process function by analyzing required samples from laboratory- or pilot-scale batches.

5.4.5 Engineering

In the Readiness stage, the Engineering function installs and qualifies all facility, utility, process, and analytical equipment, including control systems. Engineering works with vendors to facilitate FATs at the vendor sites to confirm compliance with specifications derived from the URS, and with subsequent SATs upon delivery to the RU. The capital VMP often leverages the FATs and/or SATs to reduce the number of qualification protocol deviations or to use FAT/SAT data to meet initial qualification requirements. While some developmental studies and testing may begin with equipment or at facilities that have not completed qualification, qualification of all new equipment and facility modifications must be complete and must pass all requirements prior to proceeding to the Execution stage.

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The Engineering function writes and executes all qualification protocols for facility and utility modifications, as they must be completed prior to performing PPQ batches. In addition, Engineering will provide training on specific unit operations and facility and/or utility modifications for maintenance, manufacturing, and technical staff. Whenever possible, SU and vendor personnel should participate in this training.

By the end of the Readiness stage and using input from Manufacturing, Engineering incorporates a preventive maintenance plan and spare parts requirements into the RU systems.

5.4.6 Manufacturing

The Manufacturing function writes all SOPs, work instructions, and batch records required for the RU to execute the process. These documents are used to train all personnel involved in the TT at the RU and are used as part of the EHS risk assessment, which is completed during this stage.

Manufacturing cleans the plant facilities and equipment following the Engineering qualification activities, supports the TT Team during execution of various Readiness activities, and conducts testing to determine whether the plant is ready for use. Manufacturing also should finalize documentation and training requirements prior to proceeding to the next stage to ensure the Manufacturing function is fully ready to support execution of the PPQ.

5.4.7 Regulatory

During the Readiness stage, the Regulatory function begins to obtain and author documents necessary for the regulatory submission. Some common activities include:

- Assessing any new information that may influence the regulatory filing, including presubmission advice, as well as feedback from the health authority on submission of a post-approval change management protocol
- Assessing change controls
- Assessing data from any preliminary engineering runs to provide input to the Process function in finalizing the comparability strategy
- Finalizing the regulatory strategy based on any new information discovered
- Finalizing the scope of work for the authoring and review of the regulatory submission, which includes:
  - Creating and/or finalizing a detailed submission tracker for authoring, reviewing, and approving submission documents with a matrix denoting who will be responsible, accountable, consulted, and informed for providing materials, as well as the projected timing
  - Confirming the availability of report authors and report and submission reviewers
  - Ensuring the submission tracker timeline aligns with the availability of the source information
- Reviewing validation documents supporting PPQ
- Participating in development of PAI readiness plan

The Regulatory and Quality functions will discuss the expected timing of the completion of validation documents and stability data in support of the TT. As an outcome of this stage, the regulatory strategy should be assessed against any findings during the Readiness activities including process control, validation strategy, and market strategy. Regulatory will also begin to finalize the product labels during the Readiness stage.

5.4.8 Quality

During the Readiness stage, the Quality function is extensively involved with all cross-functional areas, performing compliance reviews and approving documentation involved in preparing for PPQ, including quality elements of the control strategy. Quality ensures closure of all deviations, corrective and preventive actions (CAPAs), and change controls prior to PPQ. The Quality function should develop a procedure for PPQ batch release with input from the SU. Collaborating with the SU on the PPQ

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batch-release procedure ensures alignment on the data to be reviewed and ensures clarity for any unique aspects associated with the TT process and RU site, such as use tests or electronic batch records.

In this stage, the Quality function also should establish role requirements for product release to determine if a qualified person or a responsible person with a specific knowledge base will be needed.

Working with Regulatory, the Quality function also develops a PAI readiness plan, which may be revised as additional relevant information becomes available during the Execution stage. This should include a strategy to respond to any observations found during mock inspections.

The Quality function will qualify new vendors and audit high-risk and critical material suppliers in preparation for PPQ.

5.4.9 Sourcing/Supply Chain

The Sourcing/Supply Chain function orders all materials needed for the TT that have not yet been purchased. Materials, storage, and testing requirements are set up in the RU systems to ensure release by Quality for use when needed. Equipment and material needs should align with the overall project plan and the lead times and risks established during the Planning stage (see **Section 5.2.9**).

5.4.10 Best Practices and Additional Suggestions

Table 5.4.10-1 describes additional ideas for consideration as well as best practices for the Readiness stage.

Table 5.4.10-1 Best Practices and Additional Suggestions for the Readiness Stage

Topic	Details
Use a risk management plan for risk mitigation	• Use a risk register to ensure all potential risks are addressed and/or a mitigation plan is in place before proceeding to Execution • Systematically address process risks based on lab or manufacturing data
Use a checklist	• Use a checklist detailing all key requirements to ensure overall process readiness • Create and share a regulatory submission tracker
Ensure all documents are ready and approved	• Plan adequate time to draft, review, resolve comments on, and approve documents within the project timeline and before each due date. Consider all plant-level documentation and quality/regulatory submission documentation, such as SOPs, batch records, protocols, process descriptions, specifications for product release, and stability documents • Use templates where possible for consistency, and make sure authors, reviewers, and approvers are aware of timing expectations
Ensure all materials are ready and approved for qualification in next stage	• Ensure that material shipment lead times, site systems setups, tests, and release timelines meet project needs • Identify lead times early and make receiving parties aware of timelines to test, receive, release, and set up ahead of Execution activities. Include adequate buffer • Check in with vendors regularly to eliminate surprises in shipments and receiving
Ensure all equipment is ready and validated for qualification in next stage	• Ensure that equipment shipment lead times, installation, and validation timelines meet project needs • If facility modifications or equipment installation are needed, plan and coordinate use of shared resources (personnel and equipment) and anticipate potential impact on other processes • Identify lead times early and make receiving parties aware of timeline, with an adequate buffer included, before activities will be executed • Check in with vendors regularly to eliminate surprises in shipments and receiving

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(Table 5.4.10-1 Continued)

Topic	Details
Identify all parameters of final recommended process for verification during next stage	• Ensure that the final process being recommended is ready for the Execution stage, including all batch record instructions, SOPs, training needs, and parameters for primary, secondary, and support manufacturing systems • Ensure that process development studies are adequate and that testing and equipment qualification meet the requirements of intended process, risk assessment, and procedures • Produce a data map for process data that shows all parameters requiring control, the means of control (e.g., batch record, automated control, in-process test), and the means of recording • Use the data map to enable process monitoring and to assess whether access to data in a timely manner will present challenges during the next stage
Perform engineering run on final commercial process	• If any final changes are made to the process, or development was not performed on the final intended commercial equipment, perform an engineering batch to verify the final intended process prior to proceeding to the next stage
Establish a plan for successful regulatory filing	• Plan to draft, review, and approve filing for on-time submission within the project's timeline and launch requirements. This plan should be detailed in the submission tracker and regulatory strategy documents • Plan early initiation of section authoring. Ensure that the parties responsible for authoring allow plenty of time for review and approval, aligned with overall project timeline
Minimize hand-offs between functions and keep personnel on the project throughout	• Ensure that personnel are adequate to proceed to the next stage and that loss of personnel in a later stage will not result in a gap in expertise or a hand-off between functions (e.g., different functions qualifying equipment, developing process, and performing qualification) • Avoid reliance on contractors for all project activities. If contractors are necessary, use a mix of full-time employees and contractors to ensure sufficient internalization of key process information within the RU organization • Maintain key staffing throughout project. Use the same production personnel throughout development, process verification, and commercial stage, if possible
Take sufficient time at each stage to identify key requirements	• Use stage gate, risk management tools or gap analysis, and checklists to plan for all requirements and ensure tasks are completed prior to proceeding to the next stage • Formally assess risks of any open or incomplete actions within the TT Team and steering committee to understand the potential impact if proceeding to next stage with open items • Fully investigate any process failures or results operating on the edge of process design and, as a team, identify if development needs to be revisited prior to proceeding
Ensure supporting systems are ready for new process	• Troubleshoot, test, and qualify key equipment for readiness to begin executing activities to ensure that software, change management, spare parts, cleaning, maintenance, calibration, supporting functions, and resources are ready for the new process • Test supporting systems (e.g., batch reports, release process, validated electronic systems such as SAP, and other inventory management controls) during engineering batches to avoid PPQ surprises • Ensure that testing facilities have qualified or validated the analytical methods as required and will be able to test according to the agreed-upon schedule
Hire personnel soon enough to train before they are needed	• Ensure that functional managers and the steering committee understand the needs for resource approval, hiring, and training early in the project • Hire personnel in advance of actual need to ensure time for adequate training • Involve manufacturing operators in automation-testing, SOP preparation, batch record, and automation-development to promote learning and allow them to gain valuable experience
Conduct operations training	• Finalize work instructions for the start of batch manufacturing. This activity is typically on the critical path • Ensure operators have sufficient time to train on work instructions
Establish batch release process	• Ensure alignment among Process, Quality, and Manufacturing functions on the batch release process • Address such questions as: What batch reports will be reviewed? Will intermediates be released or processed straight through? Will batch records be reviewed on the floor as completed? What forms need to be approved and how are all requirements of the change control satisfied? Are trucks scheduled if outside normal hours?

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5.5 Execution Stage

PPQ is performed in the Execution stage, during which all operational readiness and compliance requirements are completed to allow for manufacture at the RU. Materials are tested and released to perform the engineering batch and PPQ batches, as defined in the project plan and process VMP. With multiple execution activities taking place in this stage, the team may decide to perform several stage gates, rather than just one, to continually assess the results from Execution stage activities (see **Section 5.5.1**).

Table 5.5-1 shows the activities and deliverables for the Execution stage, as excerpted from the Reference Guide (**Figure 5.0-1**). One major deliverable of the Execution stage is the submission of the regulatory filing. After the PPQ batches are complete, the TT Team usually will provide support for both the regulatory filing and the expected regulatory inspections.

5.5.1 Business

The MAH Business function is actively engaged in defining the launch strategy, anticipating completion of PPQ batches, regulatory submission, and PAI. Business will work closely with the Sourcing/Supply Chain function to define the use of the PPQ batches and estimated market demand.

For the Execution stage, multiple stage gates may be used. A stage gate may be performed prior to the engineering run to verify readiness. Another may be conducted before PPQ to verify results of all engineering and process readiness activities. Stage gates also may be executed after the engineering batch and PPQ if any changes to the process occur, risks are identified that must be mitigated, or additional studies are required. After the steering committee approves the regulatory filing, the Business function can proceed to the Handover stage.

5.5.2 Technology Transfer Project Manager

During the Execution stage, the TT PM supports the team to remove any remaining obstacles for the engineering and PPQ batches. The TT PM ensures that all documentation is approved and complies with all operational readiness requirements. All relevant change controls and deviations should be closed, and training should be completed.

As part of Execution, the TT PM ensures that all compliance and operational requirements have been met to perform PPQ with the highest probability of success. Continual communication across all organizational levels is essential. The TT PM also guides the SU and RU teams to define and execute a person-in-plant program in which SU personnel are present to support RU execution activities and associated training.

A key deliverable for the TT PM in the Execution stage is approval of the TT report documenting completion of the TT requirements defined in the TT protocol.

5.5.3 Process

The RU Process function oversees qualification exercises including shipping validation and PPQ runs. Their responsibilities include investigation of process-related deviations, execution of supporting studies as defined in the process VMP, and troubleshooting. During this stage, the Process function will provide process expertise and technical support for equipment to ensure the process is performing according to planned expectations, evaluating the effectiveness of the control strategy. Throughout the PPQ, having at least one Process staff person present for each shift of operation is recommended. Prior to starting the PPQ, the Process function ensures that the knowledge gained during the Readiness stage is incorporated clearly into the production batch records, procedures, and personnel training.

During this stage, the SU and RU Process functions should analyze data obtained during PPQ to confirm that the outcomes meet expectations. Any significant differences, as defined in the acceptance criteria and plan, should be investigated.

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Table 5.5-1 Activities and Deliverables in the Execution Stage

STAGE	Business	TT PM	Process	Analytical	Engineering	Manufacturing	Regulatory	Quality	Sourcing/Supply Chain
Execution	***Activities***								
	Develop Launch Strategy	Ensure All Good Manufacturing Practices Documentation Is Complete	Oversee Production Runs	Perform Materials Testing	Support Process Performance Qualification (PPQ)	Perform Demo/ Engineering Batch	Coordinate Regulatory Filing	Assess Deviations/ CAPAs	Manage Product Storage Requirements Prior To Release
		Participate in PAI Audit	Investigate Deviations	Perform In-Process Testing	Write Deviations	Execute PPQ Batches		Review Materials Data	
			Gather/Analyze Process Data	Perform Batch Release Testing		Execute Cleaning from PPQ		Review Batch Data	
			Perform Shipping Validation	Perform Cleaning Testing				Review Documentation	
				Package Stability Samples				Lead Execution of PAI Readiness Plan	
				Test Demo/ Engineering Batch					
	Deliverables								
	Stage Gate Approval	Equivalency Study	Process Validation Report, VMP Closure	Approved Test Results	Deviation Reports/ CAPAs	Executed Batch Records	Submit Regulatory Filing	Materials Release	Prepare Release & Distribution Plan
	Approval for Regulatory Filing	TT Report	Cleaning Validation Report	Stability Program		Clean Equipment		Product Release	
	Launch Strategy		Deviation Reports/ CAPAs			Store Product		Deviation/CAPA Resolution	
			Cycle Time, Yield					Validation Documents Approval	
			Shipping Validation Report					Close Change Controls	
								PAI Approval	

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If possible, investigations of process deviations should include members of both the SU and RU across multiple functional areas. For example, Analytical may be needed to determine whether a deviation of an in-process sample result is due to analytical performance or an issue relating to the Manufacturing function. Where genuine process failures are identified, Process and Quality should consult with Regulatory to evaluate the impact of the failure on the status of the PPQ, including parameters under which the PPQ can proceed. The TT PM should immediately communicate any failures to the steering committee.

Any necessary improvements to the process following the Execution stage should be identified before the handover to the RU post-transfer operations team. The case study in **Section 8.9** illustrates the importance of implementing process improvements during the Execution stage, prior to Handover, and the potential consequences of failing to do so.

After completion of PPQ, the Process function closes all deviations, gathers required data, and issues the process, shipping, and cleaning validation reports. Process also may be a lead author of the CMC sections of the regulatory filing.

5.5.4 Analytical

In the Execution stage, the Analytical function is in full operation, performing testing on all materials for release, in-process samples, finished products, and cleaning samples. Utilities and environmental testing may be required as well. Typically, PPQ requires more testing than is necessary during routine commercial production where a reduced testing program may be implemented provided all PPQ requirements defined in the VMPs have been met.

The Analytical function also has oversight of the packaging of stability samples as defined in the stability protocol.

5.5.5 Engineering

Having completed all qualification requirements for the facility and equipment, the Engineering function acts in a supporting role in addressing any issues that occur during the engineering and PPQ runs. Engineering will investigate deviations and issue change controls related to the equipment or facilities. Engineering may support validation-related sampling, data analysis, and reporting to satisfy all requirements of the PPQ and VMP and to ensure all requirements of the PPQ have been met.

5.5.6 Manufacturing

A major deliverable of the Execution stage for the Manufacturing function is execution of the engineering and PPQ batches. At a minimum, the SU team should participate with the RU team during the engineering batch if possible. Knowledge gained from the engineering batch commonly leads to process adjustments and improvements, prompting updates to SOPs and the master batch record. The engineering run should use materials from the suppliers that will be used for PPQ.

The engineering batch may be produced under GMP or non-GMP conditions. If conducted under GMP requirements, the engineering batch may be used to produce product for continued clinical supply. If allowed by both the process VMP and the PPQ protocol, both of which have been approved in advance, an engineering batch run under GMP conditions also may serve as a PPQ batch, particularly when the raw material supply is limited.

The Manufacturing function is a major contributor of feedback on operations, training effectiveness, and batch records following completion of the qualification activities. This feedback should be incorporated into lessons learned prior to handover to the RU post-transfer operations team.

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5.5.7 Regulatory

During the Execution stage, the Regulatory function is actively drafting the regulatory submission and, based on the submission tracker, actively managing the authoring, reviewing, and finalizing of documents. The submission timing and content should be aligned with the regulatory strategy that was finalized in the Readiness stage. The RU should provide frequent updates to the MAH Regulatory function to ensure that new and evolving information discovered during Execution can be incorporated into the submission. The RU should inform the MAH Regulatory function of the assumptions being made and of their evaluation of regulatory risks.

At the end of the Execution stage, after ensuring that the CMC content has been extensively reviewed cross-functionally, the steering committee will authorize filing of the submission. The steering committee should be apprised of risks and assumptions before authorizing filing.

The Regulatory function also works with Quality on PAI preparation. This may include suggesting high-risk items to include in storyboard development as well as ensuring alignment of the key claims made in the dossier, plant-level documentation, and storyboards. High-risk items and topics likely to arise during PAI include activities associated with deviations during PPQ and subjects that the inspectorate has historically identified as focus areas for the site, manufacturing process, product class, or company.

5.5.8 Quality

The Quality function is integral throughout the Execution stage as it releases materials for use. Quality is responsible for product release, including batch record review and overall batch disposition, in accordance with site quality procedures. Quality also is responsible for supporting the review and approval of all documentation for the PPQ, including protocols, reports, batch records, forms, results, change controls, and deviation and investigation close-outs. The Quality function would be expected to close out the quality risk assessment associated with the TT at the end of the PPQ.

At this point in the TT, Quality performs the final inspection readiness activities in anticipation of the regulatory filing which, depending on TT type, may trigger a PAI. A mock inspection (in-depth audit) of the RU performed in anticipation of a PAI may be used to identify issues that should be addressed prior to filing a submission.

Completion of the PAI may be an endpoint to the Execution stage depending on the overall business and regulatory strategies.

5.5.9 Sourcing/Supply Chain

During the Execution stage, the Sourcing/Supply Chain function will work closely with the Business function to finalize shipping requirements. The Sourcing/Supply Chain function also will work with the Manufacturing and Engineering functions on PPQ requirements, including scheduling, material, and storage needs as well as logistics. As outputs of the Execution stage, Sourcing/Supply Chain will have established suppliers, a clear supply chain, and defined standards for production planning in order to move forward to commercial manufacture.

5.5.10 Best Practices and Additional Suggestions

Table 5.5.10-1 describes best practices and additional ideas that may be considered for the Execution stage.

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Table 5.5.10-1 Best Practices and Additional Suggestions for the Execution Stage

Topic	Details
Create a technical support plan for qualification	• Ensure SMEs are on the floor prior to Execution to provide training, to support personnel should any technical challenges arise, to ensure KT from SU to RU, and to provide quick availability to resolve issues or escalations in a timely manner to prevent delay to schedule • Involve technical personnel during qualification for technical support of new equipment if issues arise and to monitor for future improvements
Ensure the team involved in qualification is trained adequately prior to the first engineering run	• Ensure that technical representatives and SMEs from the TT Team provide formal training on any new equipment and processes to the production personnel who will execute qualification batches • Ensure sufficient time to train personnel if technology or process steps are new or otherwise present a high risk for human error
Perform a trial run	• Conduct a mock or trial run with a water batch or rejected material to ensure equipment readiness and adequacy of operations training • Ensure the production team has sufficient time to practice using new equipment, parameters, batch record, and process steps before using qualification material • Provide opportunities to ask questions, improve training, incorporate process improvements and lessons learned, and reduce any human error on new process
Call a readiness meeting before running qualification batches	• Ensure the production team is clear on process requirements prior to execution • Review and/or update the control strategy • Avoid introducing new variables not challenged in previous stages • Plan ahead and reference best practices on the engineering run, in pre-batch readiness meetings, and during operator training • Use risk assessment to address any new variables that cannot be avoided
Prepare filing and plan for questions from regulatory authorities	• Provide Regulatory with documents that support the filing as they become available, in alignment with the submission tracker • Begin rolling review of drafted sections of regulatory filings as sections are (nearly) completed to manage resources and avoid unexpected delays due to reviewer availability • Identify and prepare SMEs to address questions received from regulatory authorities after initial submission
Ensure adequate development testing from the previous stage	• Provide robust supporting data and rationale from the previous stage through adequate testing to support recommended parameters, process, and instructions • Ensure a repeatable process that is controlled within its recommended operating parameters
Provide adequate batch record instructions	• Ensure all requirements and expectations are clearly defined in the batch record, and that training has been conducted and documented • Use the engineering batch to help identify any missing instructions that can be incorporated prior to the start of qualification
Ensure adequate training of production personnel	• Ensure hands-on training of personnel on the post-transfer operations team. Operator qualification is recommended over read-only training
Ensure process performs as expected within requirements and control strategy	• Ensure no changes are made to the final process prior to qualification without first revisiting the Readiness stage or performing an engineering run • Reference best practices for risk management and use the stage gate to review any uncertainties in final intended process to determine if process is ready for qualification • Review the control strategy and ensure all variables are adequately controlled with sufficient monitoring, sampling, and other mechanisms • Monitor data from engineering and PPQ batches and assess against the control strategy. At a minimum, this review should include critical parameters. Consider reviewing all parameters that meet the risk threshold • Reduce sources of variability where possible, e.g., people, materials, components

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(Table 5.5.10-1 Continued)

Topic	Details
Hold stage gates before proceeding to next phase	• Hold a stage gate meeting with the TT Team upon completion of qualification to verify the Execution stage meets all requirements and expectations, and to verify readiness to proceed to the next stage (handover to post-transfer operations team) • Ensure that all functions are aligned and that the process meets all requirements, is well controlled and repeatable, and is ready for Handover • Review any process improvements/learnings from qualification before Handover • Review risk management plan and appropriateness of mitigations

5.6 Handover Stage

The Handover stage comprises the transition of the product from the TT Team to the RU post-transfer operations team, covering routine production of marketed products. Because the risk assessment is a living document, risk identification and risk mitigation strategies should be constantly reviewed throughout the TT project. The Handover stage includes multiple activities across several functions at the RU site to ensure it is ready to work independently for commercial production or for the next stage of product development. Requirements for a smooth transition will vary depending on the type of TT and the endpoints established in the TT charter. For instance, the extent of Sourcing/Supply Chain requirements will vary based on how quickly the Business function is able to establish the customer base and volume requirements.

Ensuring that all lessons learned from the TT are incorporated into the existing documentation for improved commercial production is a key element of the Handover stage. (This technical report does not address such standard post-transfer requirements as equipment/facility maintenance, APR, life-cycle validation, complaint handling, or CPV programs.)

Table 5.6-1 shows the activities and deliverables described in the Reference Guide for the Handover stage. A stage gate is held with the steering committee and RU site management to approve formal handover of the process to the RU for routine operations.

5.6.1 Business

Once the PPQ is completed, the Business function typically performs a review of the business contract with the RU. The Business function ensures estimated commercial volumes can be produced and develops a resource plan for any changes needed to accommodate the product and its associated requirements.

5.6.2 Technology Transfer Project Manager

The TT PM ensures that the TT charter elements have been addressed in executing the TT. As part of the Handover stage, the TT PM documents all lessons learned and may help the RU incorporate key improvements prior to the final transfer. The TT PM also will ensure that the project is formally closed out and that all project requirements have been met. **Section 8.7** presents a case study of the benefits of planning ahead for commercial readiness and conducting the handover.

5.6.3 Process

The Process function of the TT Team may continue to support the product following qualification during routine post-transfer production, as Process personnel are critical to KT during the Handover stage. The Process function also may take the lead in executing process improvements identified from the PPQ, including associated change controls.

5.6.4 Analytical

Besides identifying lessons learned related to analytical activities, the Analytical function ensures that the new product is set up in the RU's stability-testing program with oversight and execution of the

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Table 5.6-1 Activities and Deliverables in the Handover Stage

STAGE	Business	TT PM	Process	Analytical	Engineering	Manufacturing	Regulatory	Quality	Sourcing/Supply Chain
Handover	***Activities***								
	Determine Changes Needed for Commercial Production	Coordinate Knowledge Transfer to Post-Transfer Operations Team	Set up Continued Process Validation (CPV)	Maintain Stability Program	Define Changeover Requirements for Future Production	Knowledge Transfer to Post-Transfer Operations Team	Obtain Regulatory Approval	Monitor Stability Data	Validation Batch Distribution
		Document Lessons Learned in TT	Document Potential Process Improvements			Maintain Validation Batch Storage	Consider Post-Approval Changes	Set up Quality Management Review Requirements	Coordinate Transportation Strategy
	Deliverables								
	Commercial Resourcing to Implement Changes	Full Handover of Product	CPV Program, Report Template (or initial report when sufficient batches produced)	Stability Results/ Report	Product Changeover Plan	Trained Personnel	Regulatory Approval	Commercial QMS	
	Stage Gate Approval							Lead Quality Management Reviews	
								Commercial Quality Agreement	

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stability protocol. Analytical also investigates out-of-specification events and deviations that may arise from stability testing.

5.6.5 Engineering

During the Handover stage, the Engineering function should consider any lessons learned relating to equipment operation during qualification. Engineering may identify process improvements, upgrades, or replacements of equipment associated with the new process. These changes will be evaluated against the site change control and filing requirements for impact prior to incorporating ahead of commercial start.

5.6.6 Manufacturing

The Manufacturing function is key to providing feedback for lessons learned and potential improvement areas following qualification. This feedback typically focuses on training, equipment operation, and the manufacturing batch record. Manufacturing will ensure all personnel supporting the ongoing commercial needs are trained adequately and are ready to support commercial demand.

Manufacturing may support and/or take ownership of CAPAs resulting from qualification that need to be addressed ahead of commercial production, such as deviations resulting from unclear instructions. The RU Manufacturing function should verify that it is ready to take on all responsibilities for commercial start, including cleaning, product changeovers, sample pulling and submission, media fills, change control, maintenance activities, and batch operations.

The case study in **Section 8.12** describes what could go wrong during commercial production if the Handover process with the Manufacturing and Sourcing/Supply Chain functions is inadequate to ensure readiness for post-transfer production.

5.6.7 Regulatory

The Regulatory function communicates expected timing for regulatory approval. Any potential delays should be communicated to the TT Team in a timely manner. Regulatory also contributes to the launch of products, reviewing promotional materials and medical information for consistency with product labeling.

The MAH Regulatory function will need to identify the RU personnel who will be responsible for ongoing communications on regulatory matters.

The TT Team may review opportunities for efficiencies or cost reductions that may result in post-approval submissions.

5.6.8 Quality

As part of the Handover stage, the MAH Quality function establishes the quality management review process, which should involve both the RU and the SU. The quality agreement may need to be updated to address everything associated with commercial manufacture.

Quality will ensure the closure of all documentation from the TT, including change controls, validation reports, deviations, and CAPAs. It also may play a role in identifying lessons learned from the Execution stage that should be incorporated in future training for the post-transfer operations team to improve process quality.

5.6.9 Sourcing/Supply Chain

The Sourcing/Supply Chain function continues to perform production planning and material sourcing activities similar to those conducted during the TT process, incorporating any improvements identified from the Execution stage.

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Following qualification, any lessons learned should be implemented prior to handover to the RU post-transfer operations team. These might include equipment and personnel standards used in schedule planning, procurement lead times and quantities, and resource planning. The supply plan and procurement plan should take into consideration commercial requirements and release expectations in accordance with business needs following qualification activities.

5.6.10 Best Practices and Additional Suggestions

Table 5.6.10-1 describes additional ideas for consideration as well as best practices for the Handover stage.

Table 5.6.10-1 Best Practices and Additional Suggestions for the Handover Stage

Topic	Details
Host a formal Handover meeting	• Hold formal a meeting between the TT Team and the post-transfer operations team to ensure overall readiness of process • Provide a Handover package including key documents, process understanding, and learnings • Provide the regulatory dossier and any critical knowledge to the commercial regulatory group and communicate any post-marketing commitments made to regulatory authorities • Have the TT Team provide an overview of new equipment, unique requirements, launch timeline, commercial forecast, labor requirements, procurement needs, open actions from risk assessment, process improvement, and lessons learned • If applicable, have the TT Team provide customer introductions • Be aware of potential issues: Without an adequate handover, additional risks may arise if the post-transfer operations team is unable to execute production effectively
Create a formal long-term training plan for production	• Prior to Handover, assess whether the production team needs additional training on the process and equipment • Use final lessons learned to revise and maintain the training program, which should include adequate hands-on training with batch records and SOPs, not only classroom presentation • Consider use of demonstrated hands-on training to verify understanding • Be aware of potential issues: Without proper training, the team runs the risk of requiring additional technical support or potentially experiencing human error, deviations, downtime, or delays upon commercial start
Perform a lessons-learned meeting	• Set up a formal meeting to discuss lessons learned at project closure, inviting the entire TT Team and involving key supporting team members (e.g., production, labs) • Discuss successes as well as opportunities for improvement, considering such areas as processes, people, communication, and documentation • Determine root causes of any challenging areas, obstacles, failures, scope changes, and deviations and discuss how to prevent them in the future • Capture a summary of lessons learned that the TT PM can drive to the rest of the team. The SU, RU, and MAH should incorporate these learnings into future projects where possible
Ensure all commercial needs are planned and ready for commercial start	• Verify that the post-transfer operations team has everything needed to begin commercial production • Work with the key functions at the RU site prior to closure of the project to ensure that key systems of the post-transfer production plan are in place and needs are met • Include such key functions as site software systems, inventory, procurement plan, resource/labor needs, scheduling standards and equipment capacity needs, spare parts, preventive maintenance, equipment requalification and routine monitoring, vendor/supplier needs, and forecasting per the commercial needs • Ensure all key aspects are ready, including procurement of materials, spare parts, lead times, scheduling, and third-party planning
Transfer knowledge from TT Team to post-transfer operations team	• Ensure formal training, robust batch-record updates, SOP instruction revisions, and simulation runs as needed are completed prior to commercial start • Ensure that knowledge has been effectively transferred through experience gained during the TT process, through lessons learned, and through Handover meetings

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(Table 5.6.10-1 Continued)

Topic	Details
Ensure capacity meets commercial needs	• Throughout the project, monitor the commercial forecast and actual cycle times on equipment, including commercial, laboratory, and any support systems at the RU site, to ensure the final process can meet the intended project and commercial requirements • If this product shares personnel and/or equipment with other products at the site, consider the impacts on equipment capacity, labor capacity, or support (e.g., lab testing, materials) • Ensure that the TT Team assesses the product's long-term needs throughout the project, involves appropriate commercial personnel and leadership early, and evaluates the commercial forecast to verify requirements are met in the lead-up to launch and Handover • Ensure all lessons learned from the project are considered in planning for commercial needs
Ensure process turned over performs as expected and is well controlled within specifications/QTPP	• Ensure an adequate process qualification that meets all acceptance criteria is well controlled within the operating ranges specified in the control strategy and is supported by sufficient data • Ensure that robust monitoring and a control plan are in place to monitor for any out-of-control events that may occur and require attention

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Note: Specific information related to Quality Risk Management for Technical Transfers is being developed as part of the Technical Report QRM series. Additional information on this topic presented in the original, now retired, version of Technical Report No. 65 can be found at PDA.org. After the issuance of the QRM series TR on this topic the retired version of Technical Report. No. 65 will be removed.

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8.0 Appendix: Case Studies

The case studies in this appendix illustrate how the technology transfer (TT) process presented in PDA *Technical Report No. 65 (Revised 2021)* can be used and how the "Reference Guide to Technology Transfer Activities and Deliverables" can be helpful in preparing for TTs. These examples of events that occurred in real TTs illustrate the benefits of using the Reference Guide and best practices. Each case study notes the relevant product lifecycle phase, TT stage, and site type. Specific regulations mentioned in individual case studies may have been updated since this TR was written. Current regulations should be consulted before undertaking any TT activity.

8.1 Technology Transfer Charter, Technology Transfer Governance, and Communication Plan

Product Lifecycle Phase: *Commercial-Commercial*
TT Process Stage: *Planning*
Site Types: *Internal-External, External-External*

To reduce costs, Company A made a business decision to consolidate outsourcing of drug product packaging, expecting to achieve savings quickly. Company A awarded a multiproduct bundle to a contract manufacturing organization (CMO), Company B, transferring packaging for 15 drug products from other contract providers. These products were new to the facility of Company B, the receiving unit (RU). This external-external TT was performed over a two-year period. The TT goal was to complete the transfer of the first product in seven months, with a new product in place each month thereafter, requiring all TTs to occur in parallel.

As Company B worked to achieve the TT goals, Company A continuously changed knowledge requirements, including labels, component design, preferred vendors and suppliers, and the qualification of backup sources. In addition, Company A asked the TT team to change the planned order of products being transferred. These changes prevented Company B's TT Team from meeting project timelines, a situation Company A escalated to senior management in both companies.

With senior leadership from both companies involved, a formal process for the TT was implemented. This process included steering committee and governance oversight, a communication plan, additional staffing, and improved TT project manager (PM) tools to allow greater visibility of the project's progress and risks. The changes specifically included the following:

- Additional resources were provided to enable key functions to focus on their specific workstream for a cluster of products based on process similarities and timeline.
- Following an all-day workshop, workstream meetings were held weekly to allow each function from both Company A and Company B to maintain alignment of their activities across the entire portfolio.
- The TT PM implemented a weekly progress meeting across workstreams during which the project timeline, needs, and risks were discussed and reported to Company A. The use of improved visual tools for these updates allowed for better alignment with project goals.
- A steering committee of one-level-higher management from both companies was formed. A weekly steering committee meeting allowed timely feedback, risk management, and escalation support from both Company A and Company B.
- Using input from the steering committee, the senior management of both companies met monthly and held face-to-face meetings on a quarterly basis throughout the project to provide overall governance.

Company B learned the value of establishing a TT charter and of ensuring that Planning stage activities were completed before moving to the next stage. After implementing the governance structure and a communication plan, Company B encountered significantly fewer change requests from Company A, allowing the TT Team to complete knowledge transfer (KT).

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Two events jeopardized the TT timeline, but the new timeline risks were quickly communicated to the steering committee for risk mitigation and the information relayed to the governance team. First, a vendor delay was resolved by negotiating a payment to the vendor to expedite the material. Second, a competing priority arose with another commercial activity using shared equipment. With the involvement of the steering committee, the TT Team was able to reach alignment on managing changes to the project scope and ensure this project received priority in order to meet launch needs.

8.2 Selecting the Right Team to Ensure Success of the Technology Transfer Project

Product Lifecycle Phase: *Development-Commercial*
TT Process Stages: *Initiation, Planning*
Site Types: *Internal-External*

Company A decided to transfer a new sterile drug product (Product X) to a CMO (Company B). Company A's limited TT experience and limited internal expertise to address the cross-functional requirements, comprising only two executive company leaders from the Quality and Supply functions, had caused problems with a previous TT.

Company B assigned dedicated personnel from cross-functional disciplines including Project Management, Account Management, Quality, Validation, Process Engineering, Planning, and Analytical. Company A realized it was understaffed for this new project and could not support these functions adequately. In the early stages of the project, Company A relied heavily on Company B's expertise to manage the project and make all decisions.

After experiencing challenges due to its understaffing in the early stages of the TT of Product X, Company A took action during the Readiness stage. With the experience gained from both in-house and external manufacturing, Company A added staff to its TT Team. It added expertise from key functions that would be required to support the new process, with a TT Team at both the marketing authorization holder (MAH) and the CMO.

As a result of Company A restructuring its team with a subject matter expert (SME) from each key function, matching each function of Company B, the project was executed more efficiently, knowledge was transferred more effectively, and communications were improved. With the enhanced internal involvement, Company A's executive team was able to provide much-needed governance and empower its TT Team to fully manage the project efficiently.

8.3 Importance of the Initiation and Planning Stages to Technology Transfer

Product Lifecycle Phase: *Commercial-Commercial*
TT Process Stages: *Initiation, Planning*
Site Types: *Internal-Internal (Regional)*

Company A needed to increase capacity for a drug substance (Product X). Since its sending unit (SU) site was unable to meet the full volume forecast of Product X, Company A made a business decision to transfer the process to another facility within its company network to serve as a second source of supply. Implementing a cloned process at the RU facility with minimal changes was critical to its regulatory strategy. To do this, the RU was instructed to purchase identical equipment (where possible) and replicate the existing process from the SU site. The SU shared full details of the equipment model and specifications, process documentation, and learnings with the RU to facilitate setup of the process.

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Because of the process equivalency required and the ability to easily share knowledge between the sites, Company A assessed this TT to be of low risk and rapid execution, with minimal development and technical requirements. As a result, Company A underestimated both the budget and the amount of time needed for completion. Additionally, the project was staffed with only part-time internal support, primarily personnel from the RU site, that did not include the cross-functional expertise needed for the TT.

Despite having almost identical equipment and process-operating history, Company A experienced multiple technical challenges. For one, Company A could not purchase the exact models from the same vendors and had to select newer models in their place, yet no risk assessment was performed to identify the impact of what seemed to be minor differences in the new equipment. As the process would be identical, a full gap assessment for the process impact at the RU was not performed. Also, a lack of resourcing for the TT at the SU site prevented a thorough KT from the SU to the RU. Subsequently, the RU could not leverage shared learnings, historical data, or expertise the SU site had gained over the years or from the original development and qualification. As a result, the process validation and operational robustness depended on dedicated SU resources to continue assistance with their historical knowledge of the process.

Internal TT teams often fail to perform adequate due diligence of a TT in the Initiation and Planning stages of the process. This case study demonstrates the importance of following a TT process, resourcing appropriately, and maintaining effective KT even for internal TTs.

8.4 Use of Regulatory Strategy for Initiation Stage Decisions

Product Lifecycle Phase: *Commercial-Commercial*
TT Process Stage: *Initiation*
Site Types: *External-External*

Company A used CMO C for manufacture of a veterinary drug product (Product Y) marketed only in the United States. When CMO C notified Company A that it no longer wanted to be a CMO, but would be manufacturing only for itself, Company A had to find a new CMO within a short time frame. Product Y is a sterile, lyophilized product with an approved indication only for the veterinary market. The TT was to include complete product and process transfer as well as analytical testing (release and stability) and stability storage.

Finding a company that can manufacture a lyophilized product can be a challenge, but this case presented additional barriers. Many companies with lyophilization capabilities manufacture lyophilized products for the Brazilian market. At the time, the Brazilian agency, ANVISA, specified that if a drug substance had no approved human indication, the veterinary product must be manufactured on a separate production line in a separate building from products for human use. Therefore, Company A needed to find a CMO that could manufacture Product Y on a separate production line from human products. As many CMOs had only one or two lyophilization production lines, the available pool of CMOs for transfer was limited.

Company A formed an initial TT Team comprising a PM and representatives from the Quality, Regulatory, and Supply Chain functions. The team reviewed the capabilities of multiple CMOs and found three companies (CMOs D, E, and F) that were able to manufacture Product Y in the U.S. in a manner that would be compliant with the current ANVISA requirements. The TT Team evaluated the CMOs based on their quality system and documentation, capability and expertise in lyophilized product manufacturing, equipment equivalence, analytical capabilities, costs, and projected timelines to commercial launch. Site evaluations revealed that CMO D was not interested due to the small

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volume that is common for veterinary products. The remaining two sites estimated similar costs, but the projected timelines to launch differed by six months, with CMO F having the longer timeline. The TT Team further investigated the drastic difference and determined that the longer timeline was more realistic. CMO F had more experience and better understood the complexities of transferring a lyophilized product. The TT Team chose CMO F, despite the longer timeline, because it had considered in more detail what needed to be done and it employed a strong PM to help guide the process.

8.5 Knowledge Transfer to Ensure Process Qualification Success

Product Lifecycle Phase: *Development-Commercial*
TT Process Stage: *Knowledge Transfer*
Site Types: *Internal-External*

Company A decided to outsource production of a new vaccine drug product. The selected CMO, Company B, had significant experience performing TTs of new processes, working with dozens of customers each year at their facility. Company A was relatively new and had limited experience in the pharmaceutical industry; in fact, this would be its first experience taking a product in its pipeline through to commercialization.

Company B had implemented lessons learned from a previous project with a different company, Company C. Company B determined that multiple delays during the project with Company C had resulted from added equipment requirements, lack of clearly established goals and boundaries, and lack of transparency during KT. Additionally, complex differences in operations had required alignment between Company B and Company C to ship and receive materials from each other, resolve unique labeling requirements, and enhance material inspection requirements.

With these learnings in mind, Company B implemented significant improvements in its TT process to ensure a successful and proactive KT with Company A. Company B developed several tools and checklists to help with the KT upon project award, including:

- Use of a handover form and checklist to establish all product requirements, TT boundaries, and goals.
- KT checklist and a prepared playbook to be shared with Company A immediately upon award, ahead of the kick-off, detailing the customer's expectations, key next steps, and key data the RU would request. The materials included a questionnaire and checklist to help both companies ask the right questions and share key documents early in the process.
- Break-out meetings for each function, held virtually prior to the TT kick-off, to ensure the kick-off meeting focused on the most impactful topics.
- A face-to-face kick-off to formally start the project, align on requirements for each phase of the project, and align on any differences already discovered through the playbook, checklist, and questionnaire.

In addition:

- The team performed an in-depth technical risk assessment of the TT to aid in effective KT for implementation at Company B, using a tracking tool and risk management document during the Execution stage to capture all knowledge and risk mitigation.
- The team leveraged on-the-floor support from key experts at Company A to capture tacit knowledge transfer between functions as the TT progressed to development personnel and again to commercial production personnel.

Use of a detailed knowledge transfer process with transparent communications allowed the TT project to be completed on time with no delays.

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8.6 Performing Process Fit Assessment to Facilitate Technology Transfer

Product Lifecycle Phase: *Development-Commercial*
TT Process Stage: *Initiation, Planning*
Site Types: *Internal-Internal*

Company X developed a multivalent vaccine (Product Alpha) and produced phase I and II clinical drug substance materials at pilot scale. Company X determined that it required a commercial facility to produce drug substance for phase III clinical trials and commercial supply. A review of the market requirements pointed to the need for a change in scale, and company policy encouraged an increase in the level of automation to improve digitization of process data, reduce reliance on paper records, and reduce potential for human error.

In order to guide equipment specifications, process engineers assigned to the TT Team at Company X (the SU) developed scaling factors for the pilot-to-commercial volumes covering each unit operation. The fact that Product Alpha was multivalent presented complications, as some serotypes had different volume requirements for the same unit operation. To avoid purchasing a piece of equipment for each serotype, process engineers grouped the serotypes according to maximum and minimum volume requirements and determined the minimum amount of process equipment required to cover all cases.

An initial process fit assessment was conducted in the Initiation stage with a general idea of requirements, followed by a detailed assessment during the Knowledge Transfer stage. With yield information from the SU, and the previously mentioned scaling factor, the detailed process fit assessment was more accurate and complete. In it, ingredient additions, flowrates, maximum and minimum vessel volumes, and similar mass balance parameters were assessed to check the fit of the process in both new and existing equipment.

It was also necessary to determine what process development work, if any, would be required to support the process transfer to the new facility, especially since a change in scale was necessary. Residence and mixing times were altered in equipment and piping in the larger facility. Such changes were considered as part of the TT risk assessment. In this case, the mixing for a key reaction step was studied using computational flow dynamics modeling and a model chemical reaction to ensure adequate mixing at the larger scale.

The process fit assessment begun during the Initiation stage and revised during the Knowledge Transfer stage was vital to ensuring that the right equipment was specified and procured and that the right process development work was conducted to ensure a successful TT of Product Alpha.

8.7 Planning Ahead for On Time Commercial Launch

Product Lifecycle Phase: *Commercial-Commercial*
TT Process Stages: *Initiation, Planning, Execution, Handover*
Site Types: *Internal-External*

Business considerations are usually the driving factor for performing a TT, with identification of the RU being the first key deliverable. In this case study, Company A (the SU) wanted to consider transferring a sterile parenteral drug product to another facility.

A small team of experts from Company A's project management, business, quality, and manufacturing functional areas performed due diligence in the choice of the RU (Company B). Once the RU was identified and a nondisclosure agreement was executed, the TT PM at the SU expanded the SU TT Team to include the Analytical, Regulatory, and Engineering functions. The RU also formed a TT Team of corresponding functional areas. These two TT teams established regular meetings to fa-

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cilitate KT and communication. Working together, the teams developed the project plan, defined the regulatory strategy for approval of the drug product, and executed a risk assessment of the knowledge elements needed to adapt the manufacturing process to the equipment at the RU in order to produce product to the quality target product profile and specifications required.

While completing the regulatory filing submission and health authority inspections, preparations began to hand over the product to manufacturing. The RU TT Team held meetings of increasing frequency closer to the start date for commercial production. During these meetings, the RU TT Team shared all lessons learned from product performance, equipment and process knowledge, and supplier management, including component and material suppliers with lead times. Formal introduction to key contacts were made, and key commercial systems such as material and batch management, supply management, and equipment were set up and ready for the new product. The RU TT Team recommended essential elements for the operator training, including lessons learned, that needed to be incorporated into the batch record. The RU TT Team also recommended changes to the production schedule to take into consideration learnings from run performance and resource needs. In addition, the RU TT Team ensured that spare parts and component needs were set up and ready for the anticipated forecast and ensured the production schedule maintained an up-to-date, 18-month forecast.

Changeover (cleaning validation) and validation of the aseptic manufacturing process at the vendor facility was on a critical timeline for manufacturing of regulatory batches, regulatory approval, and launch of commercial product from the vendor site. The TT was closed by the handover from the TT Team to a post-transfer operations team with complementary expertise and responsibilities for control of commercial production.

As a result of this successful planning and management of Handover activities, the site was ready to begin commercial production at the scheduled start without any delays.

8.8 Navigating Process, Capital, & Ownership Changes During a Drug Product Transfer

Product Lifecycle Phase: *Development-Commercial*
TT Process Stage: *All: Initiation, Planning, Knowledge Transfer, Readiness, Execution, Handover*
Site Types: *External-External*

In today's manufacturing, it is common when transferring technology from one manufacturing site to another manufacturing site for both sites to be external to the MAH. So-called "virtual companies" may start with one external supplier, then later add suppliers or switch suppliers for quality or pricing reasons. This case study, which involved the decision to switch manufacturers upon completion of clinical phase III, discusses the basics that should be considered for a seamless transfer.

Initiation: Following a capacity analysis aligned to the business strategy, the MAH decided to continue with external manufacturing. With regulatory considerations the driving factor, the MAH decided to transfer production from the external party manufacturing the clinical phase III supplies (Company A) to a different external party for commercial production. The MAH's existing cross-functional TT Team remained with the product to facilitate this new TT, and a TT proposal was drafted.

Led by the TT PM, the TT Team visited the prospective commercial site (Company B) to evaluate the potential fit and identify the process, equipment, and analytical method gaps that would need to be addressed for a successful transfer. A material supply agreement and related quality agreement were reached. Because the MAH did not want Company A to know about the pending transfer to Company B, the material supply agreement remained confidential. In this case, the MAH acted as the

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responsible SU and no communication was allowed between Company A and Company B. With the approval of the TT proposal, the steering committee made the decision to launch with Company B.

Planning: A detailed project plan that spelled out clear deliverables by phase and stage gate and detailed the project governance that was created and approved by the SU and the RU (Company B).

The same raw material vendors would be used, though all vendors needed to be requalified. For qualification, each vendor was required to supply three lots of raw materials used in manufacturing, followed by reissuing of all specifications, methods, and other supporting chemistry documents. The batch record was transferred, and the equipment was purchased to match as closely as possible the current process. In this case, the scale was to be increased, which resulted in a new registration batch followed by three validation batches. Based on process knowledge and guided by the process failure mode effects analysis, a process validation master plan was approved that defined the necessary protocols to be executed.

The MAH discussed scalable processes, engineering batches, and process sampling and controls. The decision was made to separate the batch record into two—one batch record for mixing and one for filling—as there was a significant hold time after mixing and a check was needed on the product before filling. The steering committee approved the stage gate, authorizing the TT Team to proceed to full-scale manufacturing.

Execution: Once new equipment was installed and validated, dynamic smoke studies were conducted, and Company B decided to perform media fills using their specific vial and equipment to demonstrate that aseptic conditions were maintained under the slightly changed conditions. Microbiological testing of all personnel and the environment was performed throughout the process using touch plates, and a swab study was performed after a risk assessment to determine the appropriate environmental monitoring program.

According to the stability plan, stability studies were performed in both the upright and inverted positions for all batches. All the RU's cleaning procedures, cleaning swab-testing, protocols, reports, training, and risk assessments were determined to be adequate as is and were used for the transfer. Serialization, packaging, and labeling were agreed upon. Company B wanted the equipment to be able to label for unit-dosing before unit-dose labeling became mandatory.

Methods were successfully transferred and were qualified or validated per requirements.

Project Closure/Lessons Learned: The TT was successful but, in hindsight, the process could have been smoother had the MAH and Company B identified early on which party would be responsible for the purchase of the additional capital. In the end, the MAH and the RU agreed to share these costs, with the exception that the Company B agreed to purchase a couple of the more expensive items. An additional difficulty was that the MAH had not negotiated or retained the rights to the methods, batch records, and all documents necessary for the transfer that Company A was performing for them. Had the MAH preemptively ensured it owned those rights, the TT would have been easier to execute.

8.9 Failed Process Qualification Due to Incomplete Process Readiness & Knowledge Transfer

Product Lifecycle Phase: *Commercial-Commercial*
TT Process Stage: *Readiness, Execution*
Site Types: *External-External*

Company A made a business decision to outsource its manufacturing and packaging process for Product X, a new vaccine. At the time, Company A outsourced Product X with a CMO, Company B, but

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was unhappy with their services. Based on cost savings and strategy, Company A decided to discontinue business with Company B by the end of the year, and it awarded a TT project to a new CMO (Company C) with an aggressive timeline. Company A informed Company C that the product must be successfully launched within 12 months to ensure that commercial supply was not interrupted. Company A stressed this criticality, as they could not go back to Company B for additional commercial business.

Company A and Company C kicked off the project smoothly and progressed to Planning and Readiness activities. Upon execution of the engineering runs, however, results were suboptimal. Company C had several ideas to improve the process to better control the results, ensure higher yield, improve confidence in right-first-time process qualification and, from their experience, avoid longer-term out-of-specification results during commercial production. These improvements would require additional, unbudgeted spending by Company A to perform additional batches, however, as well as additional time, which would push the overall timeline beyond the 12 months specified and make the target launch impossible.

Company A and Company C were misaligned and escalated the issues to their managements, losing several weeks in the process. Company C felt strongly that it should not proceed with process qualification with the current suboptimal results, and Company A wanted to proceed, at risk, to meet supply needs. Finally, a compromise was reached: Company C's TT Team agreed to make minor improvements to the current process with current equipment, capabilities, batch records, and trainings, and to conduct one additional engineering run to improve process understanding and verify improvements. If those steps went well, the team would move into process qualification. Assuming those batches passed all requirements, they would be commercially saleable. Following release of these initial launch batches, the Company C TT Team would revisit the proposed improvements. Company A accepted liability should the strategy of moving ahead with the qualification without implementing all identified improvements fail.

Unfortunately, the process qualification batch did not pass. Three months were lost executing the initial improvements and engineering run, analyzing data and investigating the root cause of the failure, finalizing a recommended path forward to improve the process, and obtaining alignment on additional costs and an adjusted timeline. The team agreed on a path forward and Company C successfully launched the product by the middle of the following year, almost six months late. Company A was forced to consider extending business relations with Company B to supply their product in the interim or otherwise impact market supply.

8.10 Effective Use of Knowledge Transfer

Product Lifecycle Phase: *Commercial-Commercial*
TT Process Stage: *Initiation, Planning, Execution*
Site Types: *External-Internal*

In this case study, the technology of a commercialized sterile drug product (mAb-X) approved for commercial drug product manufacturing at an external site (Site A) was transferred to Site B, the RU. Site B was an internal site of the MAH for mAb-X (Company Y) and was new to commercial drug manufacturing. Site A and Site B did not communicate directly during the TT. The Manufacturing Science and Technology group of Company Y served as the SU and was responsible for the KT from Site A to the Site B.

Company Y had experience with TT for product mAb-X having previously transferred the product to Site A in a development-to-commercial phase TT. Therefore, Company Y already possessed information important to the TT, such as the commercial manufacturing process, required equipment,

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process parameters classification, risk assessment for the commercial process, and analytical methods for product quality testing. This information was used as a reference for the TT of the same product to Site B.

The goal of the TT was to minimize the number of changes to the existing manufacturing process used by Site A while designing a process for commercial manufacture that fit within the constraints of Site B. To achieve this objective, a facility-fit and gap assessment was performed for Site B, which became the key to the success of the TT. The existing risk assessment information for mAb-X at Site A was used as the starting point for the risk assessment for site B. Also, since the RU was a new commercial drug product manufacturing site, the TT team worked with the on-site Validation group at Site B to map out the equipment validation activities required to support this TT (installation qualification, operational qualification, and performance qualification). An integrated project plan was drafted during the Planning stage that included an overview of all TT activities needed for the successful TT and their interdependencies.

The SU shared the information on analytical methods from the prior mAb-X TT with the RU. An evaluation for implementation of existing analytical methods for Site B was completed, and resources were allocated for the method transfer, training, and execution. Change controls and documentation procedures for manufacturing mAb-X were established before the TT activities to Site B were executed. All changes related to process, raw material specifications, equipment, and process parameters for the TT were captured under the central change control management system.

As part of the Readiness stage, the sourcing of raw materials, components, and equipment was reviewed, and relevant information was included in the validated electronic system (in this case, SAP). The information for product, packaging, and component specifications was reviewed and approved by both the RU and the SU. In addition, upon request, vendors/suppliers supplied drawings and other required information. Suppliers also provided certificate of analysis information for all raw materials and components, which was reviewed. Inspection and acceptance criteria for receiving raw materials and components in a GMP facility were agreed upon by the RU and SU teams with input from the Quality function.

As part of process performance qualification (PPQ) readiness activities, the SU Engineering group, in collaboration with their RU counterparts, prepared process description documents capturing the commercial manufacturing process for mAb-X at Site B. Information from the process description document for mAb-X at Site A was used as a template, and appropriate changes were made to capture the differences in the process at Site B. Process characterization studies were carried out to support the changes in the commercial process for Site B based on the facility-fit and gap assessment exercise, which was critical for the success of this TT. Also, prior knowledge from the TT of mAb-X to Site A was leveraged for process characterization studies.

Engineering runs were performed using surrogate and using mAb-X to understand the design space for the operation of process parameters and to determine the normal operating range and proven acceptable range for various unit operations. The required number of PPQ batches were manufactured using a matrix approach to bracket the batch size (minimum and maximum), and the rationale for the PPQ strategy was documented prior to execution. Post-PPQ, all deviations were closed and documented, and required CAPAs were identified and implemented. At this point, PPQ summary reports were written that captured the process parameter data for all PPQ batches, hold-time data for various process steps, product quality attribute results (in-process and lot-release testing results), and deviations for the PPQ batches. In addition, at the end of the PPQ campaign, completed batch records and cleaning batch records were reviewed.

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8.11 Use of Stage Gates to Address Process Readiness Challenges

Product Lifecycle Phase: *Development-Commercial*
TT Process Stage: *Readiness*
Site Types: *Internal-Internal (Global)*

Company A transferred a very complex process involving a new drug product from development (Product X) in this case study. The new molecular entity required new technologies for which limited industry expertise existed. The TT Team encountered significant delays, and they spent more than two years troubleshooting various problems with the new equipment and repeating developmental studies on the new equipment to finalize the process parameters and develop a robust process. The TT Team did not feel it was possible to meet the company leadership's goals of both advancing Product X to PPQ to meet launch requirements and maintaining the project within the adjusted budget, which was well over the original budget. This misalignment prompted the TT Team to implement a steering committee and stage gates.

The team designed the formal TT process to allow both functional management and senior leadership to review key project data, risks, accomplishments, and proposed options for each phase of the project for risk mitigation. By engaging senior leaders from elsewhere within the company, senior management was able to provide additional expertise on the new technology and similar processes. These leaders also offered different perspectives on the suboptimal data. Following review of the data, and with support from management to improve the process, the TT team agreed on the next steps, tried the recommended strategies, and leveraged on-site support from proposed SMEs within the company. The improvements to the processes resulted in a product that met requirements.

Before proceeding, the TT Team facilitated another stage gate meeting to review recent data and ensure alignment on next steps. The process was much improved, but given the unique process and requirements, new technologies, and limited expertise, the steering committee decided the TT Team should execute one more study to facilitate the learnings and gain additional experience. That study would be followed by a mock run (water batch) to improve the training and experience of the production personnel prior to proceeding to PPQ. Ultimately, the team successfully executed their qualification batches right first time. Although the added developmental studies and mock run took additional time, the TT Team agreed that by managing the previous misalignment, they had saved time by preventing a failed qualification or Handover.

8.12 Importance of Handover Stage for a Successful Commercial Start

Product Lifecycle Phase: *Development-Commercial*
TT Process Stage: *Handover*
Site Types: *Internal-Internal*

Company A had little experience with TT; however, with a new building and equipment, it had gained the technological capability to support a new sterile drug product transfer being planned (Product X). In addition, Company A hired personnel, bringing new functions into the facility including TT PMs, scientists, regulatory affairs staff, and process engineers. Company A also brought in several experienced consultants to support the new equipment and its validation at the facility, who stayed on through the Product X launch. The TT Team was well supported with project staffing, regular stage gates, a steering committee, and engaged leadership. Due to their limited experience in TT and the risks associated with the new process, equipment, and technology, the company management also ensured that SMEs would be closely involved to monitor and support each step of the project.

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During development and process qualification, the TT Team used an "all hands on deck" approach, expecting the entire team to participate on the floor and oversee the process during execution of the studies and commercial batches. The team managed the checklists, oversaw the batch records, and led each step to ensure the production operators and area personnel were correctly executing the process requirements. Production personnel also incorporated interventions, adjustments, and recommendations based on learnings from the TT Team's experience. The team wanted to make sure that the expertise and learnings from their history with the project and equipment were leveraged to ensure the commercial batches would be completed right first time. As a result, the project was successful: the commercial batches were executed flawlessly and met all requirements for release. Following project closure, the TT Team personnel were repurposed to other positions within the company and the consultant contracts were terminated.

As the facility entered routine commercial operations for this product, however, the production personnel found the new process and product requirements very challenging. Without the expertise on the floor to oversee and guide them every step of the way, the batch records, SOPs, and training proved insufficient. Due to the production personnel's lack of understanding on how to set up and operate the new equipment, the facility experienced significant downtime and eventually had to obtain onsite vendor support to retrain and create training guides for the team. Because of the lost time, failed batches due to human error, investigations, and CAPAs, the manufacturing team decided to update the SOPs and batch records associated with the new process and retrain the personnel. After several months—and significant product losses and costs—the production personnel had obtained the necessary knowledge on the process and had received adequate training through the updated SOPs and batch records to successfully and repeatably execute the processes without additional support.

In hindsight, if the TT Reference Guide approach had been used, critical knowledge elements could have been identified in the KT stage, and risk mitigation strategies could have been developed to ensure the successful deployment of commercial operations.

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